

07/12/2022

Numerical Methods:-

$$\begin{array}{r} 2 \overline{) 28,0} \\ 2 \overline{) 14,0} \\ 2 \overline{) 7,1} \\ 2 \overline{) 3,1} \\ 1 \end{array} \quad \begin{array}{c} (11100)_2 \\ \boxed{00011100} \\ 8 \text{ bit} \end{array}$$

0011100.1

$$0.5 \times 2 = 1.0$$

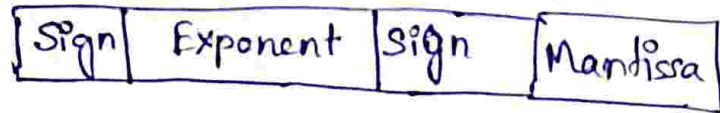
$$1.0 \times 2 = 2.0$$

285

$$-2^{n+1}, -2^{n-1}$$

$$-2^{\frac{n}{2}}$$

- Unsigned int $\rightarrow$ only integer e.g: 28
- Signed int $\rightarrow$ sign integer e.g: -28
- Unsigned fixed points $\rightarrow$ Integer Fraction e.g: 28.5
- Signed fixed points $\rightarrow$ sign integer Fraction e.g: -28.5, -43.625
- Floating point no. representation



e.g. ~~43.625~~

$$\begin{array}{r} 2 \overline{) 43,1} \\ 2 \overline{) 21,1} \\ 2 \overline{) 10,0} \\ 2 \overline{) 5,1} \\ 2 \overline{) 2,0} \\ 1 \end{array}$$

$$0.625 \times 2 = 1.250$$

$$0.250 \times 2 = 0.500$$

$$0.500 \times 2 = 1.000$$

-43.625

$$= \boxed{11101011.101}$$

32 bit representation

1 bit sign

1

15 bit integer

000000000101011

16 bit fraction.

1010000000000000

• Floating point representation



-53.5 $0.5 \times 2 = 1.0$

$$\begin{array}{r} 2 \overline{) 53.1} \\ 2 \overline{) 26.0} \\ 2 \overline{) 13.0} \\ 2 \overline{) 6.0} \\ 2 \overline{) 3.0} \\ \hline 1 \end{array}$$

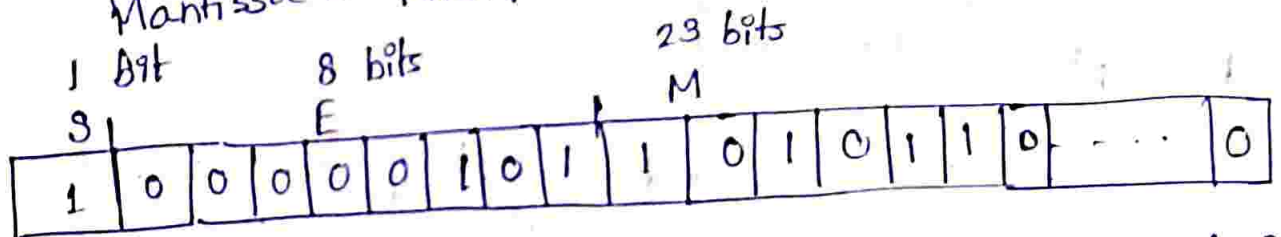
$\boxed{1110101.1} = -53.5$

$= -(110101.1)_2 \rightarrow$ denormalised form

$= -(1.101011) \times 2^5 \rightarrow$ normalised form

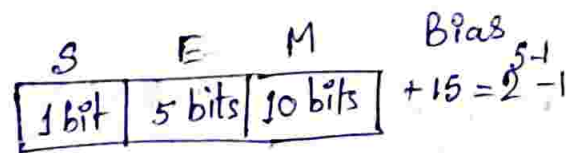
Exponent = 5 = 101

Mantissa = 101011

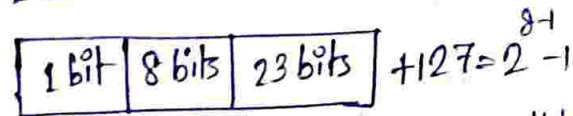


• IEEE-754 Standard the floating point no. is represented in following ways:

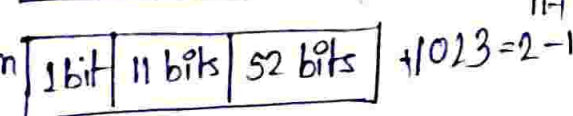
Half precision 16 bit representation



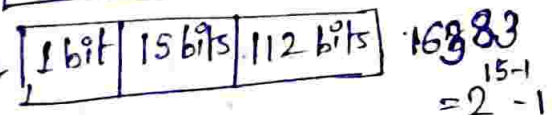
Single precision 32 bit representation



Double precision 64 bit representation



Quadruple precision 128 bit representation



-2^{n-1} to 2^{n-1}

9/12/2022

Numerical Methods:-

① IEEE 754 - 32 bit -

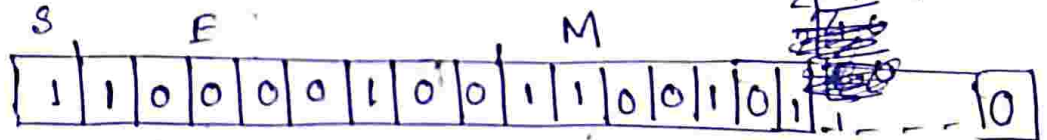
$$-57.25 = -(111001.01)_2$$

$$\begin{array}{r} 2 \overline{) 57.1} \\ 2 \overline{) 28.0} \\ 2 \overline{) 14.0} \\ 2 \overline{) 7.1} \\ 2 \overline{) 3.1} \\ 1 \end{array}$$

$$= -(111001.01)_2 \times 2^5$$

$$\text{Exponent} = 5 = 101$$

$$\text{Mantissa} = 1100101$$



$$0.25 \times 2 = 0.50$$

$$0.50 \times 2 = 1.00$$

$$5 + 127 = 132$$

$$2 \overline{) 132.0} = 1000100$$

$$\begin{array}{r} 2 \overline{) 66.0} \\ 2 \overline{) 33.1} \\ 2 \overline{) 16.0} \\ 2 \overline{) 8.0} \\ 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 4.0} \\ 2 \overline{) 2.0} \\ 1 \end{array}$$

② IEEE 754 (64 bit) : $-(237.625)_{10}$

$$\begin{array}{r} 2 \overline{) 237.1} \\ 2 \overline{) 118.0} \\ 2 \overline{) 59.1} \\ 2 \overline{) 29.1} \\ 2 \overline{) 14.0} \\ 2 \overline{) 7.1} \\ 2 \overline{) 3.1} \\ 1 \end{array}$$

$$237 = 11101101$$

~~0.6~~

$$-(237.625)_{10}$$

$$= -(11101101.101)_2$$

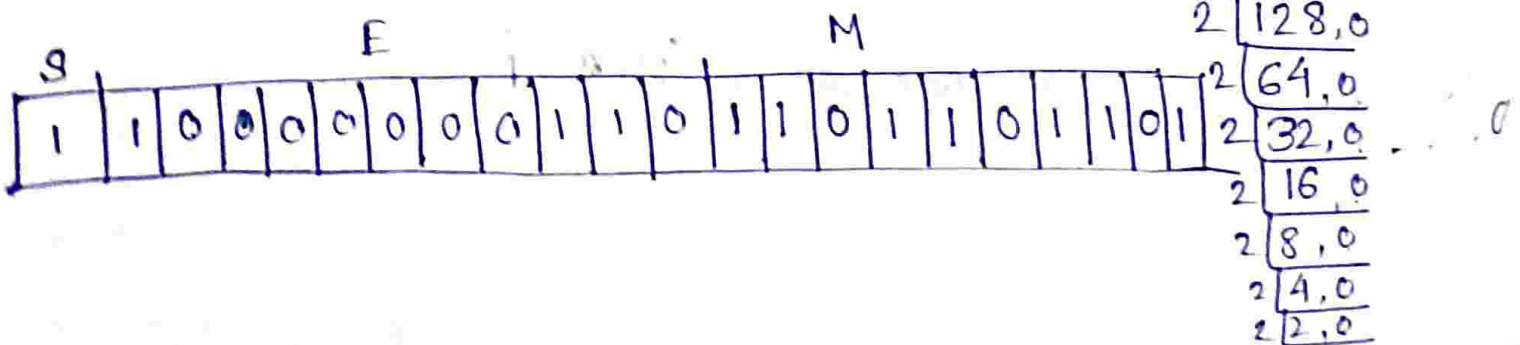
$$= -(1.1101101101)_2 \times 2^7$$

$$0.625 \times 2 = 1.250$$

$$0.250 \times 2 = 0.50$$

$$0.50 \times 2 = 1.00$$

$$7 + 1023 = 1030$$



$$\begin{array}{r} 2 \overline{) 1030.0} \\ 2 \overline{) 515.1} \\ 2 \overline{) 257.1} \\ 2 \overline{) 128.0} \\ 2 \overline{) 64.0} \\ 2 \overline{) 32.0} \\ 2 \overline{) 16.0} \\ 2 \overline{) 8.0} \\ 2 \overline{) 4.0} \\ 2 \overline{) 2.0} \end{array}$$

Why do we study NM?

- function is a mapping (relation) from domain to range.

- What is pixel?

- How is a display made? Analog/

- Why do we see 3D but camera sees 2D?

Digital Camera

Space is limited in RAM so we use floating point and fixed point representation so that data does not exceed the space allotted.

B.P

173
175
178
180
182

888

14
29

36
177.6
5 | 888
5

38
35

38
35

30
30

x

Limitation of memory space

Secondary memory

→ RAM
much smaller than secondary memory

Representation

Fixed point

Floating point

32bit

64bit

16/12/2022 Numerical Methods:-

SOLUTION OF EQUATION:-

• Polynomial equation:-

$\rightarrow f(x) = 0 \Rightarrow f(x_i) = 0$ where x_i satisfies $f(x) = 0$ for $i = 1, 2, \dots, n$

$\Rightarrow f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n = 0$ (Algebraic equation)

n roots of $f(x)$ as highest power of x is n .

$\rightarrow x e^{2x} - 1 = 0$ or $\cos x - x e^x = 0$ (Transcendental equation)
[Exponential/Trigonometric Functions]

$\rightarrow$ Root: A no. α , for which $f(\alpha) = 0$ is called a root or zero of the equation $f(x) = 0$.

$\rightarrow$ Types of Roots:-

• Simple root:- A no. α is a simple root if $f(\alpha) = 0$
if $f(\alpha) = 0$ and $f'(\alpha) \neq 0$.

e.g. $f(x) = x^3 + x - 2 = 0$

$\therefore x - 1 = 0$ or $x^2 + x + 2 = 0$

~~$\Rightarrow x^2 + 2x - x - 2 = 0$~~

$\therefore f(x) = 0 \Rightarrow x = 1$

~~$\Rightarrow x(x+2) - 1(x+2) = 0$~~

(simple root)

$\Rightarrow x^3 - x^2 + x^2 - x + 2x - 2 = 0$

$\Rightarrow x^2(x-1) + x(x-1) + 2(x-1) = 0$

$\Rightarrow (x-1)(x^2 + x + 2) = 0$

$$g(x) = x - 1 = 0$$

$$g'(x) = 1 (\neq 0)$$

$$g'(1) = 1 (\neq 0)$$

$$\text{Also, } f(x) = x^3 + x - 2$$

$$\Rightarrow f'(x) = 3x^2 + 1$$

$$\Rightarrow f'(1) = 3 + 1 = 4 (\neq 0)$$

$$f(x) = g(x)h(x)$$

$$\Rightarrow f'(x) = g'(x)h(x) + h'(x)g(x)$$

• Multiple roots: A no. α is a multiple root of multiplicity m , of $f(x) = 0$ if $f(\alpha) = 0, f'(\alpha) = 0, \dots, f^{(m-1)}(\alpha) = 0, f^{(m)}(\alpha) \neq 0$.

Here, $f(x) = (x - \alpha)^m g(x), g(\alpha) \neq 0$. $[g(\alpha) = 0$ will contradict the condition of multiplicity]

$$\text{e.g. } f(x) = x^3 - 3x^2 + 4 = 0$$

$$\Rightarrow x^3 + x^2 - 4x^2 - 4x + 4x + 4 = 0$$

$$\Rightarrow x^2(x+1) - 4x(x+1) + 4(x+1) = 0$$

$$\Rightarrow (x+1)(x^2 - 4x + 4) = 0$$

$$\Rightarrow (x+1)(x-2)^2 = 0$$

$\Rightarrow 2$ is a root with multiplicity 2.

$$f(x) = x^3 - 3x^2 + 4 \Rightarrow f(2) = 2^3 - 3 \cdot 2^2 + 4 = 8 + 4 - 12 = 0$$

$$f'(x) = 3x^2 - 6x \Rightarrow f'(2) = 3 \cdot 2^2 - 6 \cdot 2 = 12 - 12 = 0$$

$$f''(x) = 6x - 6 \Rightarrow f''(2) = 12 - 6 = 6 \neq 0$$

→ Polynomial equation of degree $n \Rightarrow$ exactly n roots.

- real
- complex
- simple
- multiple

→ Transcendental equation $\Rightarrow$ it may have 1 root or infinitely many roots or no roots.

→ Methods of finding roots:-

- Direct method
- Indirect method Iterative method

→ Quadratic equation:- $ax^2 + bx + c = 0, a \neq 0$.

$$\text{solution: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \begin{matrix} \text{[direct solution]} \\ \text{by} \end{matrix}$$

* Iteration means using move one step and using it move to the next step.

e.g. for loop: $\text{for } i = 1 \text{ to } 10 \quad (10 \text{ steps})$
 $\quad \quad \quad j = j + 1 \quad \text{start when } i = 1 \text{ [initialisation]}$
 $\quad \quad \quad \text{print } j \quad \text{stop when } i = 10 \text{ [stopping criterion]}$

→ Iterative solution:- → Start with initial approximation.

→ Obtain a sequence of approximations.

→ $x_1, x_2, \dots, x_k, \dots \quad k \rightarrow \infty$

$$f(x_k) = 0$$

An iterative method for finding root of the equation $f(x)=0$

$$x_{k+1} = \phi(x_k), \quad k = 0, 1, 2, \dots \quad \text{when } x$$

We stop when $|f(x) - \hat{f}(x)| < \epsilon$ i.e., $f(x) = 0$ is the solution

$$\text{so } |0 - \hat{f}(x)| < \epsilon$$

$$\Rightarrow |\hat{f}(x)| < \epsilon \quad [\text{Stopping criterion}] \quad \text{--- ①}$$

Also if $|x_{k+1} - x_k| < \epsilon$ [Stopping criterion] --- ②

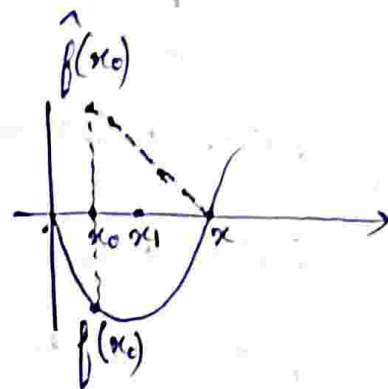
[where ϵ is the threshold and very small].

e.g. 2 decimal accuracy $|x_{k+1} - x_k| < 0.005$

3 decimal accuracy $|x_{k+1} - x_k| < 0.0005$

→ We have to calculate $x_{k+1} = \phi(x_k)$

So, $x_1 = \phi(x_0)$ initial approx.



- We approx the initial as close to the real value as possible to reduce the no. of steps.

→ For polynomial equations, Descartes's rule of sign gives the bound for the number of positive and negative real roots.

$$P_n(x) = 0 \quad \text{e.g. } 8x^3 - 12x^2 - 2x + 3 = 0 \quad (8, -12, -2, +3)$$

$$\underbrace{1}_{+} + \underbrace{1}_{+} = 2 \text{ changes in sign.}$$

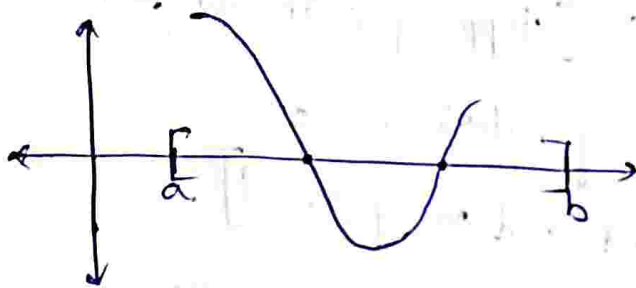
∴ Equation will have

2 positive or 0 positive roots.

$$\therefore f(-x) = -8x^3 - 12x^2 + 2x + 3 = 0 \quad (-8, -12, 2, 3)$$

∴ Equation will have, 1 negative root $\underbrace{1}_{+} = 1 \text{ change in sign}$

→ Theorem-1: If $f(x)$ is continuous on some interval $[a, b]$ and $f(a)f(b) < 0$, then the equation $f(x) = 0$ has at least one real root or an odd no. of real roots in the interval (a, b) .



e.g. ① $f(x) = 8x^3 - 12x^2 - 2x + 3 = 0$ 2 +ve; 1 -ve roots.

$$f(1) = 8 - 12 - 2 + 3 = 11 - 14 = -3$$

$$\therefore f(1)f(2) = (-3)(15)$$

$$f(2) = 64 - 48 - 4 + 3 = 67 - 52 = 15$$

$$= -45 < 0$$

$\therefore$ At least 1 root will lie between $[1, 2] \Rightarrow$ 1 positive root.

$$f(0) = 3$$

$$f(0)f(-1) = 3(-15)$$

$$f(-1) = -8 - 12 + 2 + 3 = -20 + 5 = -15$$

$$= -45 < 0$$

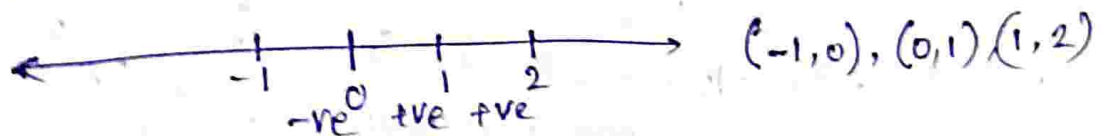
$\therefore$ At least 1 root will lie between $[-1, 0] \Rightarrow$ 1 negative root.

$$f(0) = 3$$

$$\therefore f(0)f(1) = 3(-3) = -9 < 0$$

$$f(1) = -3$$

$\therefore$ Another root will lie between $[0, 1] \Rightarrow$ 1 positive root



e.g. ⑩ $f(x) = xe^x - \cos x = 0$

$$f(0) = 0 - \cos 0 = -1$$

$$f(0)f(1) = -2.178 < 0$$

$$f(1) = 1e^1 - \cos 1 = e - \cos 1 = 2.178$$

∴ ~~At least~~ At least 1 root will lie between $[0, 1] \Rightarrow$ 1 positive root.

BOOK Numerical Methods by SRK Iyengar + R.K Jain.
Solve solved + unsolved problems.

→ Iterative Methods:-

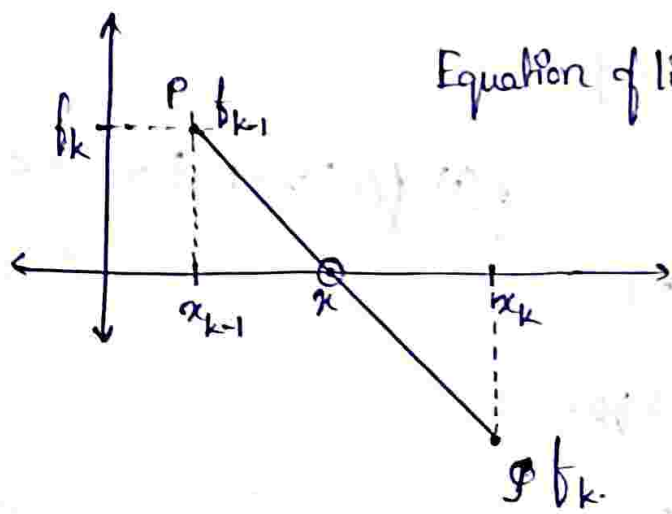
① Method of False Position:

- Linear Interpolation method
- Chord method
- Regula-falsi Method.

Q. 1. Find the interval in which the root lies. $f(x) = 0, (x_{k-1}, x_k)$

$$b_k = f(x_k) \quad b_{k-1} = f(x_{k-1})$$

$$\Rightarrow b_{k-1}b_k < 0$$



$$\text{Equation of line: } \frac{y - b_k}{b_{k-1} - b_k} = \frac{x - x_k}{x_{k-1} - x_k}$$

$$\text{At } x, y = 0$$

$$\therefore \frac{-b_k}{b_{k-1} - b_k} = \frac{x - x_k}{x_{k-1} - x_k}$$

$$\Rightarrow x = \frac{-b_k(x_{k-1} - x_k)}{b_{k-1} - b_k} + x_k$$

$$\therefore x_{k+1} = x_k - \left(\frac{x_{k-1} - x_k}{b_{k-1} - b_k} \right) f_k$$

$$x_{k+1} = \phi(x_k)$$

Pros: Convergence is confirmed as we know the interval.

Cons: One End point is always fixed so we have to do more iterations to reach the solution.

e.g. $f(x) = x^3 - 3x + 1 = 0$ $(1, -3, 1)$
 $1 + 1 = 2$ +ve roots

$f(-x) = -x^3 + 3x + 1 = 0$ $(-1, +3, +1)$
 $1 = 1$ -ve root

$$f(0) = 1$$

$$f(0) f(1) = 1(-1) = -1 < 0$$

$$f(1) = 1 - 3 + 1 = 2 - 3 = -1$$

1 positive root in $[0, 1]$

$$f(-1) = -1 + 3 + 1 = 4 - 1 = 3$$

$$f(1) f(2) = (-1)3 = -3 < 0$$

$$f(-2) = -8 + 6 + 1 = -8 + 7 = -1$$

1 negative root in $[-2, -1]$

$$f(2) = 8 - 6 + 1 = 9 - 6 = 3$$

$$f(1) f(2) = (-1)3 = -3 < 0$$

1 positive root in $[1, 2]$

Formula: $x_{k+1} = \frac{x_k - 1 f_k - x_k b_{k-1}}{b_k - b_{k-1}}$

$$x_0 = 0, x_1 = 1$$

$$x_2 = \frac{x_1 f(0) - x_0 f(1)}{f(0) - f(1)} = \frac{0 - 1}{1 - (-1)} = \frac{-1}{2} = -0.5$$

$$x_2 = \frac{0(-1) - (1)1}{-1 - 1} = \frac{-1}{-2} = 0.5 \quad f(0.5) = -0.375$$

$$f(0) = 1, f(1) = -1$$

$$f(0.5) = -0.375$$

$$x_0, x_2 \quad x_3 = \frac{x_0 f_2 - x_2 f_0}{f_2 - f_0} = 0.36364, f(x_3) = -0.04235$$

$$(0, 0.36364)$$

[Python program to implement method of false position.]

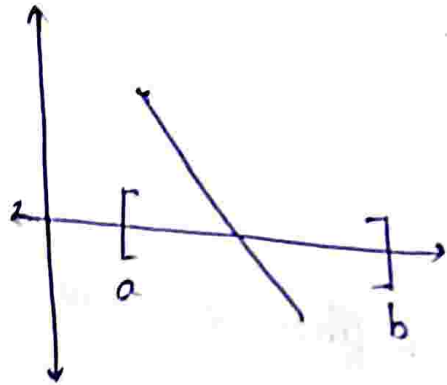
ASSIGNMENT-1 16/12/2022 - 23/12/2022

Numerical Methods:-

21/12/2022

→ $f(x) = 0$ [Method of False Position]

$[a, b], f(a) f(b) < 0 \Rightarrow$ a root in (a, b) .



• $[a, b]$

$f(x_0)$

$f(a) f(x_0) < 0 \checkmark \rightarrow$

or, $f(x_0) f(b) < 0$

• $[a, x_0]$

$f(a) f(x_1) < 0$

$\rightarrow f(x_1) f(x_0) < 0$

• $[a, x_1]$ $f(a) f(x_2) < 0$

$f(x_1) f(x_2) < 0$

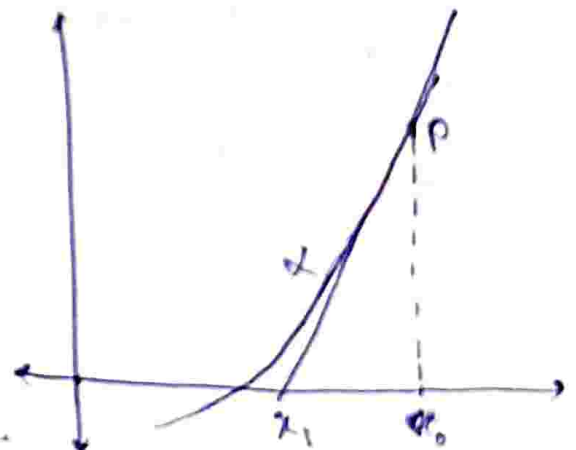
→ But this is a slow method; hence Newton-Raphson method or Chord Method was developed.

Newton-Raphson Method:-

$y = f(x), P(x_0, f(x_0))$

$$y - f(x_0) = (x - x_0) f'(x_0)$$

Slope of the tangent to the curve.



$$\frac{y - f(x_0)}{f'(x_0)} = x - x_0$$

Regula falsi → sure of
[Method of false position] convergence

$$\Rightarrow x = x_0 - \frac{f(x_0)}{f'(x_0)} \quad [\text{at } y=0]$$

Newton-Raphson → faster

Assignment-2

Implement Newton Raphson method in Python. 30th Dec, 2022

• $f(x) = \frac{1}{x} - N$, $N \neq 0$. Find $1/N$ for a given $N=17$.

① $x_0 = 0.05$ ② $x_0 = 0.15$

$$f(x) = \frac{1}{x} - N \Rightarrow f'(x) = -\frac{1}{x^2}$$

③ $x_0 = 0.05$

$$f(x_0) = \frac{1}{0.05} - N \quad f'(x_0) = -\frac{1}{(0.05)^2}$$

$$= 20 - N$$

$$= -400$$

$$\therefore x_1 = 2x_0 - Nx_0^2$$

$$= 2(0.05) - 17(0.05)^2$$

$$= 0.0575$$

$$x_2 = 2x_1 - 17x_1^2 = 0.058794$$

$$x_3 = 2x_2 - 17x_2^2 = 0.058823$$

$$x_4 = 2x_3 - 17x_3^2 = 0.058823$$

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$= x_k - \frac{\frac{1}{x_k} - N}{\left[-\frac{1}{x_k^2}\right]}$$

$$= x_k + \frac{(1 - Nx_k)x_k}{1}$$

$$= x_k + x_k - Nx_k^2$$

$$= 2x_k - Nx_k^2$$

$\therefore |x_4 - x_3| = 0$, the

iterations converge to the root. The required root is 0.058823.

(vi)

$$x_0 = 0.15$$

$$\therefore x_1 = 2x_0 - 17x_0^2 = 0.825$$

$$x_2 = 2x_1 - 17x_1^2 = -0.280706$$

$$x_3 = 2x_2 - 17x_2^2 = -1.900942$$

$$x_4 = 2x_3 - 17x_3^2 = -65.23275$$

We find that $x_k \rightarrow -\infty$ as k increases.

$\therefore$ The iterations diverge very fast.

Eg. find smallest positive root for $f(x) = x^3 - 5x + 1 = 0$.

$$f(0) = 1$$

$$f(0)f(1) = -3 < 0$$

$$f(1) = 1 - 5 + 1 = -3$$

$$f'(x) = 3x^2 - 5$$

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$\text{Let } x_0 = 0.5$$

$$x_1 = \frac{2x_0^3 - 1}{3x_0^2 - 5} = 0.176471$$

$$x_2 = \frac{2x_1^3 - 1}{3x_1^2 - 5} = 0.201568$$

$$x_3 = \frac{2x_2^3 - 1}{3x_2^2 - 5} = 0.201640$$

$$x_4 = \frac{2x_3^3 - 1}{3x_3^2 - 5} = 0.201640$$

$$\begin{aligned} &= x_k - \frac{x_k^3 - 5x_k + 1}{3x_k^2 - 5} \\ &= \frac{3x_k^3 - 5x_k^3 + 5x_k^2 - x_k^3 + 5x_k - 1}{3x_k^2 - 5} \\ &= \frac{2x_k^3 - 1}{3x_k^2 - 5} \end{aligned}$$

Hence, $|x_4 - x_3| = 0 < \epsilon$, Hence, the root is 0.201640

Program Newton Rhapson:-

Input: 1. function $f(x)$ [Ask for the function to be solved]

2. Initial guess x_0 (Ask the user for initial x_0 /

Regula falsi: $[a, b]$

(Ask the user the interval, compute x_0 .

NR:

Compute $f(x)$

3. Compute interval + x_0 (NR)

Regula falsi:

NR:

Compute x_{k+1}

Compute $f'(x)$

Compute $f(a)$

Compute x_{k+1}

$f(x_{k+1})$ $f(b)$

$f(a) f(x_{k+1}) < 0$

$f(b) f(x_{k+1}) < 0$

x	-3	-2	-1	0	1	2	3
$f(x)$							

Root lies in () Check $f(a) f(b) < 0$

x_0

Output: ask for the interval

ask for x_0

30th Dec - Mid Sem.

28th Dec - Syllabus.

Numerical Methods

22/12/2022

Linear System of Algebraic Equations:-

single power, non-transcendental, more than 1 equation:-

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

;

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_m$$

System of Linear Equation.

Variables:-

x_i where $i=1, 2, \dots, n$

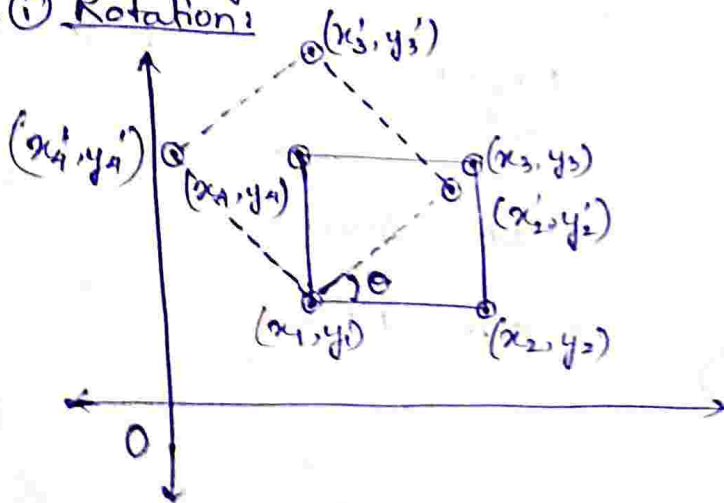
Right hand side Values (Constants)

$$b_i, \quad i = 1, 2, 3, \dots, m.$$

Co-efficients:- (Constants)

$$a_{ij}^0, i = 1, 2, 3, \dots, m$$
$$j = 1, 2, 3, \dots, n$$

e.g. (i) Rotation:



$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_1' \\ y_1' \end{bmatrix}$$

System formed after rotation
about (x_1, y_1) .

Say $\theta = 30^\circ$

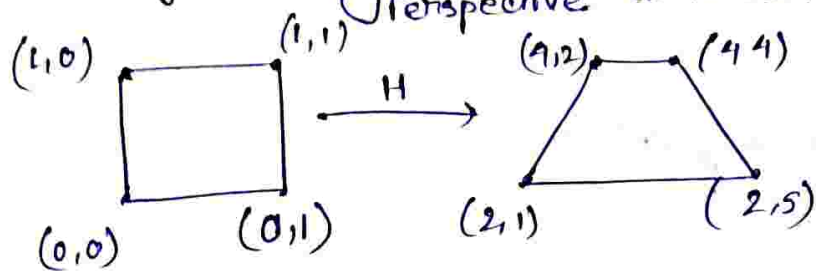
$$\therefore \cos 30^\circ x_4 - \sin 30^\circ y_4 = x'_4 = 30$$

$$\sin 30 x_A + \cos 30 y_A = y'_A = 10$$

$$\Rightarrow \left. \begin{aligned} \frac{\sqrt{3}}{2} x_1 - \frac{1}{2} y_1 &= 30 \\ \frac{1}{2} x_1 + \frac{\sqrt{3}}{2} y_1 &= 10 \end{aligned} \right\} \text{system for } (x_1, y_1)$$

$$\Rightarrow \frac{1}{2}x_4 + \frac{\sqrt{3}}{2}y_4 = 10 \quad \text{--- (1)}$$

eg. (i) Image Mapping:



$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Similarly, 3 other systems can be formed.

So we get,

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}_{m \times n} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m \times 1}$$

$\Rightarrow \boxed{Ax = b}$ matrix representation of system of linear equation

(i) A = co-efficient matrix. $[A|b]$ = augmented matrix.
 $m \times (n+1)$

(ii) System of equations $Ax = b$ is consistent (atleast one solution),

if $\text{rank}(A) = \text{rank}([A|b]) = r$. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ linearly independent.

(iii) If $\text{rank}(A) = n$ (considering only square matrices), Hence, $\text{rank} = 2$, then a unique solution exists.

(iv) If $\text{rank}(A) < n$, then the system has $(n-r)$ parameter family of infinite no. of solutions.

② (v) System of equations $Ax = b$ is inconsistent if $\text{rank}(A) \neq \text{rank}[A|b]$ (no solution)

• Solution of system of linear equations:-
 ↳ Direct Method
 ↳ Iterative Method

① Direct method:-

① → A is a diagonal matrix. $A = D = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix}_{n \times n}$ s.t. $a_{ii} \neq 0$
 $\forall i = 1, 2, \dots, n$

$\therefore Ax = b \Rightarrow Dx = b$

$\Rightarrow \begin{bmatrix} a_{11} & 0 & 0 & \dots & 0 \\ 0 & a_{22} & 0 & \dots & 0 \\ 0 & 0 & a_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & a_{nn} \end{bmatrix}_{n \times n} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}_{n \times 1}$

$\Rightarrow a_{11}x_1 = b_1 \Rightarrow x_1 = \frac{b_1}{a_{11}}$
 $a_{22}x_2 = b_2 \Rightarrow x_2 = \frac{b_2}{a_{22}}$
 $\vdots$
 $a_{nn}x_n = b_n \Rightarrow x_n = \frac{b_n}{a_{nn}}$

② → A is an upper triangular matrix. $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix}_{n \times n}$

$\therefore Ax = b \Rightarrow \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$

We back track from a_{nn} , i.e., $a_{nn}x_n = b_n$

$\Rightarrow x_n = \frac{b_n}{a_{nn}}$

then $a_{(n-1)(n-1)}x_{n-1} + a_{(n-1)n}x_n = b_{n-1}$

$\Rightarrow x_{n-1} = \frac{1}{a_{(n-1)(n-1)}} \left[b_{n-1} - a_{(n-1)n} \frac{b_n}{a_{nn}} \right]$ and so on.

$$x_1 = \frac{b_1 - \sum_{j=2}^n a_{1j} x_j}{a_{11}}$$

④ similarly, we can solve lower triangular matrix but from top to bottom.

$$Ax = b \Rightarrow \begin{bmatrix} a_{11} & 0 & 0 & \dots & 0 \\ a_{21} & a_{22} & 0 & \dots & 0 \\ a_{31} & a_{32} & a_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$$x_1 = 5$$

$$2x_2 = 7 \Rightarrow x_2 = \frac{7}{2}$$

$$4x_3 = 1 \Rightarrow x_3 = \frac{1}{4}$$

$$6x_4 = 2 \Rightarrow x_4 = \frac{1}{3}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 1 \\ 2 \end{bmatrix}$$

④ $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$ $a_{ij} \neq 0 \quad \forall i, j = 1, 2, \dots, n$.

We use row transformation to change A to any of the above mentioned forms.

① Interchange any two rows:-

$$\text{e.g. } \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & 0 & a_{23} & b_2 \\ 0 & a_{32} & a_{33} & b_3 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a_{32} & a_{33} & b_3 \\ 0 & 0 & a_{23} & b_2 \end{array} \right]$$

② Division/Multiplication of any row by a non-zero no. p.
i.e., pR_i , (i^{th} row)

© Adding/ Subtracting a scalar multiple of any row to any other row. e.g

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{R_1' = R_1 + 2R_2} \begin{bmatrix} a+2c & b+2d \\ c & d \end{bmatrix}$$

→ Similarly, we can use column transformation.

- Gauss Elimination Method:-

$$Ax = b \xrightarrow[\text{row transformation}]{\text{elementary}} Ux = z, \text{ then solve.}$$

[Here U = upper triangular matrix]

$$[A|b] \xrightarrow{\text{Gauss Elimination}} [U|z]$$

28/12/2022:- Numerical Methods:-

• Gauss Elimination:- $Ax = b \rightarrow Ux = c$, where U = upper triangular matrix.

• Gauss Jordan:- $Ax = b \rightarrow Ix = d$, where I = Identity matrix

$$[A|b] \rightarrow [I|d] \quad x = A^{-1}b \quad \text{Finding } A^{-1}$$

$$AA^{-1} = A^{-1}A = I$$

$$A_{n \times n} x_{n \times 1} = b_{n \times 1}$$

$$\Rightarrow A_{n \times n}^{-1} A_{n \times n} x_{n \times 1} = A_{n \times n}^{-1} b_{n \times 1}$$

$$\Rightarrow I_{n \times n} x_{n \times 1} = d_{n \times 1}$$

• A is a square matrix.

• A^{-1} should exist.

• $\det(A) \neq 0$

• Full Rank $\Rightarrow$ linearly independent matrix.

• $A \rightarrow I$ Elementary Row operations.

$$\textcircled{1} \quad x_1 + x_2 + x_3 = 1$$

$$4x_1 + 3x_2 - x_3 = 6$$

$$3x_1 + 5x_2 + 3x_3 = 4$$

$$1 + \frac{5}{2} = \frac{7}{2} - \frac{1}{2} = \frac{6}{2} = 3$$

$$-2 + \frac{5}{2} = -\frac{4}{2} + \frac{5}{2} = \frac{1}{2}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 4 & 3 & -1 & 6 \\ 3 & 5 & 3 & 4 \end{array} \right] \xrightarrow[R_3 - 3R_1]{R_2 - 4R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & -5 & 2 \\ 0 & 2 & 0 & 1 \end{array} \right] \xrightarrow{(-1)R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 5 & -2 \\ 0 & 2 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 5 & -2 \\ 0 & 0 & -2 & 5 \end{array} \right] \xrightarrow{(-\frac{1}{2})R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 5 & -2 \\ 0 & 0 & 1 & -\frac{5}{2} \end{array} \right] \xrightarrow{R_2 - R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{5}{2} \end{array} \right]$$

$$\xrightarrow{R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{5}{2} \end{array} \right] \xrightarrow{R_1 - R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{5}{2} \end{array} \right]$$

$$x_1 = 3, \quad x_2 = \frac{1}{2}, \quad x_3 = -\frac{5}{2}$$

$$3 + \frac{1}{2} - \frac{5}{2} = 3 + \frac{1-5}{2} = 3 - 2 = 1$$

② Inverse of A using Gauss Jordan:-

$$[A|I] \rightarrow [I|A^{-1}]$$

$$\text{e.g. } \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 4 & 3 & -1 & 0 & 1 & 0 \\ 3 & 5 & 3 & 0 & 0 & 1 \end{array} \right] \xrightarrow[R_3 - 3R_1]{R_2 - 4R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & -5 & -4 & 1 & 0 \\ 0 & 2 & 0 & -3 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{(-1)R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 5 & 4 & -1 & 0 \\ 0 & 2 & 0 & -3 & 0 & 1 \end{array} \right] \xrightarrow{R_3 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 5 & 4 & -1 & 0 \\ 0 & 0 & -10 & -11 & 2 & 1 \end{array} \right]$$

$$\xrightarrow{(-\frac{1}{10})R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 5 & 4 & -1 & 0 \\ 0 & 0 & 1 & \frac{11}{10} & -\frac{2}{10} & -\frac{1}{10} \end{array} \right] \xrightarrow[R_1 - R_3]{R_2 - 5R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & -\frac{1}{10} & \frac{2}{10} & \frac{1}{10} \\ 0 & 1 & 0 & -\frac{3}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{11}{10} & -\frac{2}{10} & -\frac{1}{10} \end{array} \right]$$

$$\xrightarrow{R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{7}{5} & \frac{1}{5} & -\frac{2}{5} \\ 0 & 1 & 0 & -\frac{3}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{11}{10} & -\frac{2}{10} & -\frac{1}{10} \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} \frac{7}{5} & \frac{1}{5} & -\frac{2}{5} \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ \frac{11}{10} & -\frac{2}{10} & -\frac{1}{10} \end{bmatrix}$$

$$1 - \frac{11}{10}$$

$$= \frac{10-11}{10}$$

$$= -\frac{1}{10}$$

$$-\frac{1}{10} + \frac{3}{2}$$

$$\begin{aligned} \frac{1}{10} - \frac{1}{2} &= \frac{-2+10}{20} \\ &= \frac{8}{20} = \frac{2}{5} \\ &= -\frac{8}{20} = -\frac{2}{5} \end{aligned}$$

$$4 - 5 \times \frac{11}{10}$$

$$= \frac{8-11}{2}$$

$$= -\frac{3}{2}$$

$$-1 + \frac{3}{2}$$

$$= -1+1$$

$$= 0$$

$$0 + \frac{9}{10} = \frac{1}{2}$$

$$AA^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{5} & \frac{1}{5} & -\frac{2}{5} \\ -\frac{3}{2} & 0 & \frac{1}{2} \\ \frac{1}{10} & -\frac{2}{10} & -\frac{1}{10} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{7}{5} - \frac{3}{2} + \frac{1}{10} & \frac{1}{5} + 0 - \frac{2}{10} & -\frac{2}{5} + \frac{1}{2} - \frac{1}{10} \\ \frac{28}{5} - \frac{9}{2} - \frac{11}{10} & \frac{4}{5} + 0 + \frac{2}{10} & -\frac{8}{5} + \frac{3}{2} + \frac{1}{10} \\ \frac{21}{5} - \frac{15}{2} + \frac{33}{10} & \frac{3}{5} + 0 - \frac{3}{10} & -\frac{6}{5} + \frac{3}{2} - \frac{3}{10} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{14-15+1}{10} & 0 & \frac{-4+5-1}{10} \\ \frac{56-45-11}{10} & \frac{5}{5} & \frac{-16+15+1}{10} \\ \frac{42-75+33}{10} & 0 & \frac{-12+25-3}{10} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{array}{r} 25 \\ -15 \\ \hline 10 \\ \hline 10 \end{array}$$

$$\frac{5-5}{10} = 0$$

$$\frac{56-56}{10} = 0$$

$$\frac{25-15}{10} = \frac{10}{10} = 1$$

Iterative Methods:-

$$Ax = b \quad x_0 = \text{initial solution}$$

$$\downarrow$$
$$\text{Final } x \quad x = A^{-1}b$$

$$x^{k+1} = A^{-1}b, \text{ such that } |x^{(k+1)} - x| < \epsilon$$

$$x^{k+1} = Mx^{(k)} + c$$

$\downarrow$
[Iteration matrix]

$$\text{Stopping criterion: } |x^{(k+1)} - x^{(k)}| < \epsilon$$

• Gauss Jacobi Method:-

$$Ax = b$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

From, 1st row, $(k+1)$

$$x_1^{(k+1)} = \frac{1}{a_{11}} [b_1 - a_{12}x_2^{(k)} - a_{13}x_3^{(k)}]$$

2nd row, $(k+1)$

$$x_2^{(k+1)} = \frac{1}{a_{22}} [b_2 - a_{21}x_1^{(k)} - a_{23}x_3^{(k)}]$$

3rd row, $(k+1)$

$$x_3^{(k+1)} = \frac{1}{a_{33}} [b_3 - a_{31}x_1^{(k)} - a_{32}x_2^{(k)}]$$

$$x_i = \frac{b_i}{a_{ii}}, \quad x_i = 0 \text{ if } i$$

• We need to start with an initial vector.

$$\begin{bmatrix} x_1^0 \\ x_2^0 \\ x_3^0 \end{bmatrix}$$

• If we guess the initial initial vector better, we can reach solution faster with fewer no. of steps otherwise the sequence of iterations might diverge.

$$\textcircled{1} 4x_1 + x_2 + x_3 = 2$$

$$x_1 + 5x_2 + 2x_3 = -6$$

$$x_1 + 2x_2 + 3x_3 = -4$$

$$x_1^{(0)} = 0, x_2^{(0)} = 0, x_3^{(0)} = 0$$

$$x_1^{(k+1)} = \frac{1}{4} [2 - x_2^{(k)} - x_3^{(k)}] \quad x_3^{(k+1)} = \frac{1}{3} [-4 - x_1^{(k)} - 2x_2^{(k)}]$$

$$x_2^{(k+1)} = \frac{1}{5} [-6 - x_1^{(k)} - 2x_3^{(k)}]$$

$$\therefore x_1^{(1)} = \frac{1}{4} [2 - x_2^{(0)} - x_3^{(0)}] \quad x_2^{(1)} = -\frac{6}{5} \quad x_3^{(1)} = -\frac{4}{3}$$

$$= \frac{1}{4} \cdot 2 = \frac{1}{2}$$

$$x_3^{(1)} = -\frac{4}{3}$$

$$x_1^{(2)} = \frac{1}{4} \left[2 + \frac{4}{3} + \frac{6}{5} \right] = \frac{1}{4} \frac{[30 + 20 + 18]}{15} = \frac{68}{4 \times 15} = \frac{17}{15}$$

$$x_2^{(2)} = \frac{1}{5} \left[-6 - \frac{1}{2} + \frac{8}{3} \right] = \frac{1}{5} \frac{[-36 - 3 + 16]}{6} = \frac{-23}{5 \times 6} = -\frac{23}{30}$$

$$x_3^{(2)} = \frac{1}{3} \left[-4 - \frac{1}{2} + \frac{12}{5} \right] = \frac{1}{3} \frac{[-40 - 5 + 24]}{10} = \frac{-21}{3 \times 10} = -\frac{7}{10}$$

and so on, till $|x^{k+1} - x^k| < \epsilon$

To reach solution faster we can do the following method:

$$x_1^{(k+1)} = \frac{1}{a_{11}} [b_1 - a_{12}x_2^{(k)} - a_{13}x_3^{(k)}]$$

$$x_2^{(k+1)} = \frac{1}{a_{22}} [b_2 - a_{21}x_1^{(k+1)} - a_{23}x_3^{(k)}]$$

$$x_3^{(k+1)} = \frac{1}{a_{33}} [b_3 - a_{31}x_1^{(k+1)} - a_{32}x_2^{(k+1)}]$$

This is Gauss Sedal Method.

This will reduce the no. of iterations for larger matrices and hence reach solution faster.

So we will take our guesses of solution as:

$$x_1^{(k+1)} \rightarrow x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)} \quad |x_1^{k+1} - x_1^k| < \epsilon_1$$

$$x_2^{(k+1)} \rightarrow x_1^{(k+1)}, x_2^{(k)}, \dots, x_n^{(k)} \quad |x_2^{k+1} - x_2^k| < \epsilon_2$$

$$\vdots \quad |x_3^{k+1} - x_3^k| < \epsilon_3$$

$$x_n^{(k+1)} \rightarrow x_1^{(k+1)}, x_2^{(k+1)}, \dots, x_{n-1}^{(k+1)}, x_n^{(k)} \quad \epsilon_1, \epsilon_2, \epsilon_3 \text{ can be same or different.}$$

e.g. $-4x_1 + x_2 - x_3 = -8$

Initial Guess:

$$3x_1 - 6x_2 + 2x_3 = 23$$

$$x_1^{(0)} = 0.9, x_2^{(0)} = -3.1, x_3^{(0)} = 0.9$$

$$x_1 - 3x_2 + 7x_3 = 17$$

$$x_1^{k+1} = -\frac{1}{4} [-8 - x_2^k + x_3^k]$$

$$x_1^1 = -\frac{1}{4} [-8 + 3.1 + 0.9]$$

$$= 1$$

$$x_2^{k+1} = \frac{-1}{6} [23 - 3x_1^{k+1} - 2x_3^k]$$

$$x_2^1 = -\frac{1}{6} [23 - 3(1) - 2(0.9)]$$

$$= -\frac{1}{6} = -3.0167$$

$$x_3^{k+1} = \frac{1}{7} [17 - x_1^{k+1} + 3x_2^{k+1}]$$

$$x_3^1 = \frac{1}{7} [17 - 1 + 3(-3.0167)]$$

$$= 0.9928$$

$$x_2^1 = -\frac{1}{4} [-8 + 3.0167 + 0.9928]$$

$$= 0.9976$$

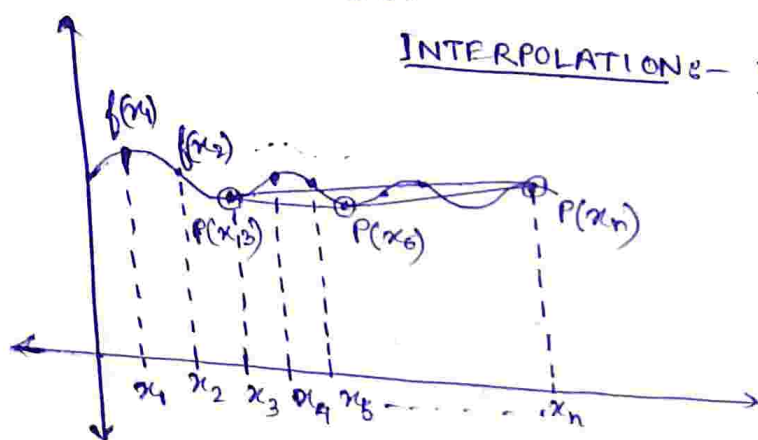
$$x_2^2 = -\frac{1}{6} [23 - 3(0.9976) - 2(0.9928)]$$

$$= -3.0036$$

$$x_3^2 = \frac{1}{7} [17 - 0.9976 + 3(-3.0036)] = 0.9988 \text{ and so on.}$$

Numerical Methods:-

19/01/2023



It is approximating a function with sampling and guessing as close to the original function as possible.

$$P(x) \approx f(x)$$

$$f(x_i), i = 1, 2, 3, \dots, n.$$

We don't know about $f(y)$, for $x_i < y < x_j$
for any i, j .

Let, $f(x)$: Continuous function defined on an interval $[a, b]$.
 $n+1$ points $x_0, x_1, \dots, x_n$ such that $x_0 < x_1 < \dots < x_n$,
 $f(x_1), f(x_2), \dots, f(x_n)$ is given.

$x_0, x_1, x_2, \dots, x_n$ equispaced (not necessary).

Polynomial approx: Find a polynomial $P_n(x)$ of degree $\leq n$ such that,

$P_n(x_i) = f(x_i) \forall i = 0, 1, 2, \dots, n$] Interpolating conditions.

$P_n(x)$ is known as the interpolating polynomial.

(Method to find the interpolation polynomial)

Lagrange Interpolation: Unevenly Spaced points:-

$x_0, x_1, x_2, \dots, x_n$ s.t. $|x_i - x_{i+1}|$ same $\forall i$.

① Use all the ordinates $\rightarrow f(x_0), f(x_1), \dots, f(x_n)$ to interpolation.

② Unique polynomial - $(n+1)$ points.

x	x_0	x_1	$\dots$	x_n
$f(x)$	$f(x_0)$	$f(x_1)$	$\dots$	$f(x_n)$

• Unevenly spaced points/non-uniformly spaced points

$x_0, x_1, \dots, x_n; (n+1)$ points.

Fit a polynomial of degree $\leq n$.

@ x_0 @ x_1

$P_1(x) \rightarrow \text{degree} \leq 1$

Polynomial will be a straight line $\rightarrow$ if 2 unique points $\rightarrow x_0, x_1$

$P_n(x) = l_0(x) f(x_0) + l_1(x) f(x_1) + \dots + l_n(x) f(x_n) \rightarrow \text{①}$

where, $f(x_0), f(x_1), \dots, f(x_n)$ are constants.

We need, $P_n(x_i) = f(x_i)$, $\forall i = 0, 1, 2, \dots, n$.

$$\therefore P_n(x_0) = f(x_0) \text{ when } l_i(x_0) = 0, \forall i = 1, 2, \dots, n \quad \text{--- (2)}$$

$$\therefore \text{from (1), } P_n(x_0) = l_0(x_0)f(x_0) \text{ [as } l_i(x_0) = 0 \forall i = 1, 2, \dots, n \text{]} \quad \text{--- (3)}$$

$\therefore$ from (2) and (3),

$$l_0(x_0) = 1$$

$$\text{So, we get, } l_i(x_j) = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

$$l_i(x) = 0 \quad ; \quad x = x_0, x_1, x_2, \dots, x_{i-1}, x_{i+1}, \dots, x_n$$

$$l_i(x) = c(x-x_0)(x-x_1)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)$$

$$l_i(x_i) = 1 = c(x_i-x_0)(x_i-x_1)\dots(x_i-x_{i-1})(x_i-x_{i+1})\dots(x_i-x_n)$$

$$\Rightarrow c = \frac{1}{(x_i-x_0)(x_i-x_1)\dots(x_i-x_{i-1})(x_i-x_{i+1})\dots(x_i-x_n)}$$

$$\therefore l_i(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)}{(x_i-x_0)(x_i-x_1)\dots(x_i-x_{i-1})(x_i-x_{i+1})\dots(x_i-x_n)}$$

$$P_n(x) = l_0(x)f(x_0) + l_1(x)f(x_1) + \dots + l_n(x)f(x_n) \text{ where, } l_i(x) \text{ is given by (5).} \quad \text{--- (5)}$$

23/01/2023.

Numerical Methods:-

$$\text{Error} = |f(x) - P(x)| < \epsilon$$

x	0	1	3
$f(x)$	0	1	0

$$n+1 = 3$$

$$n = 2$$

$$l_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-1)(x-3)}{(0-1)(0-3)} = \frac{(x-1)(x-3)}{3}$$

$$P_n(x) = l_0(x) \cancel{f(x_0)}^0 + l_1(x) \cancel{f(x_1)}^1 + l_2(x) \cancel{f(x_2)}^0$$

$$= l_1(x) f(x_1)^1$$

$$= l_1(x)$$

$$= \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}$$

$$= \frac{(x-x_0)(x-x_2)}{(1-0)(1-3)} = \frac{(x-0)(x-3)}{-2} = \frac{-x(x-3)}{2} = \frac{3x-x^2}{2}$$

$$\therefore P_n(x) = \frac{3x-x^2}{2}$$

$$P_n(0.5) = \frac{3 \cdot \frac{1}{2} - \left(\frac{1}{2}\right)^2}{2}$$

$$P_n(x_1) = f(x_1)$$

$$P_n(x_0) = \frac{3 \cdot 0 - 0^2}{2} = 0$$

$$P_n(x_1) = \frac{3 \cdot 1 - 1^2}{2} = \frac{3-1}{2} = \frac{2}{2} = 1$$

$$P_n(x_2) = \frac{3 \cdot 3 - 3^2}{2} = 0$$

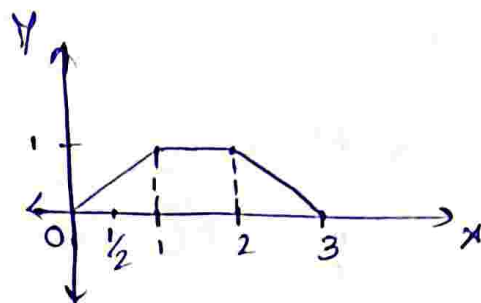
$$= \frac{\frac{3}{2} - \frac{1}{4}}{2}$$

$$= \frac{\frac{3}{4} - \frac{1}{8}}{2}$$

$$= \frac{6-1}{8}$$

$$= \frac{5}{8}$$

$$P_n(2) = \frac{3 \cdot 2 - 2^2}{2} = 3 - 2 = 1$$



① Divided difference:-

Given: Let the data be $(x_i, f(x_i))$,
 $i = 0, 1, 2, \dots, n$.

Good approximation of
 functions.

First divided difference: $(x_i, f(x_i)), (x_{i+1}, f(x_{i+1}))$.

$$f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}, \quad i = 0, 1, 2, \dots, n-1$$

$$f[x_{i+1}, x_i] = \frac{f(x_i) - f(x_{i+1})}{x_i - x_{i+1}} = \frac{-(f(x_{i+1}) - f(x_i))}{-(x_{i+1} - x_i)} = f[x_i, x_{i+1}]$$

Second divided difference:

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

$$= \frac{1}{(x_2 - x_0)} \left[\left(\frac{f(x_2) - f(x_1)}{x_2 - x_1} \right) - \left(\frac{f(x_1) - f(x_0)}{x_1 - x_0} \right) \right]$$

$$= \frac{f(x_0)}{(x_2 - x_0)(x_1 - x_0)} - \frac{f(x_1)}{(x_2 - x_0)(x_2 - x_1)} - \frac{f(x_1)}{(x_2 - x_0)(x_1 - x_0)} + \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)}$$

$$= \frac{f(x_0)}{(x_2 - x_0)(x_1 - x_0)} - \frac{f(x_1)}{(x_2 - x_0)} \left(\frac{1}{x_2 - x_1} + \frac{1}{x_1 - x_0} \right) + \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)}$$

$$\begin{aligned}
 & + \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)} \\
 &= \frac{f(x_0)}{(x_2 - x_0)(x_1 - x_0)} - \frac{f(x_1)}{(x_2 - x_0)} \frac{(x_1 - x_0 + x_2 - x_1)}{(x_2 - x_1)(x_1 - x_0)} + \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)} \\
 &= \frac{f(x_0)}{(x_2 - x_0)(x_1 - x_0)} - \frac{f(x_1)}{\cancel{(x_2 - x_0)}} \frac{(x_2 - x_0)}{(x_2 - x_1)(x_1 - x_0)} + \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)} \\
 &= \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2)} + \frac{f(x_1)}{(x_1 - x_0)(x_1 - x_2)} + \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)}
 \end{aligned}$$

ⁿth ~~Third~~ divided difference:-

$$f[x_0, x_1, x_2, \dots, x_n] = \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{(x_n - x_0)}$$

Divided difference:-

x	$f(x)$	First d.d	2nd d.d	3rd d.d	...
x_0	$f(x_0)$	$f[x_0, x_1]$			
x_1	$f(x_1)$	$f[x_1, x_2]$	$f[x_0, x_1, x_2]$		
x_2	$f(x_2)$	$f[x_2, x_3]$	$f[x_1, x_2, x_3]$	$f[x_0, x_1, x_2, x_3]$	
x_3	$f(x_3)$				
$\vdots$	$\vdots$				

Newton's Divided Difference Interpolation:-

$f(x) = P_n(x)$ polynomial of degree n such that,

$$P_n(x_i) = f(x_i) \quad \forall i$$

$$P_n(x) = f(x_0) + (x-x_0)f[x_0, x_1] + (x-x_0)(x-x_1)f[x_0, x_1, x_2] \\ + \dots + (x-x_0)(x-x_1)\dots(x-x_{n-1})f[x_0, x_1, x_2, \dots, x_n]$$

Q.

x	-4	-1	0	2	5
$f(x)$	1245	33	5	9	1335

$$P_n(x) = f(x_0) + (x-x_0)f[x_0, x_1] + (x-x_0)(x-x_1)f[x_0, x_1, x_2] \\ + (x-x_0)(x-x_1)(x-x_2)f[x_0, x_1, x_2, x_3] \\ + (x-x_0)(x-x_1)(x-x_2)(x-x_3)f[x_0, x_1, x_2, x_3, x_4]$$

Given: First divided difference:-

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{33 - 1245}{-1 + 4} = \frac{-1212}{3} = -404$$

$$f[x_1, x_2] = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{5 - 33}{0 + 1} = -28$$

$$f[x_2, x_3] = \frac{f(x_3) - f(x_2)}{x_3 - x_2} = \frac{9 - 5}{2 - 0} = \frac{4}{2} = 2$$

$$f[x_3, x_4] = \frac{f(x_4) - f(x_3)}{x_4 - x_3} = \frac{1335 - 9}{5 - 2} = \frac{1326}{3} = 442$$

Second divided difference:-

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{-28 + 404}{0 + 4} = \frac{376}{4} = 94$$

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1} = \frac{2 + 28}{2 + 1} = \frac{30}{3} = 10$$

$$f[x_2, x_3, x_4] = \frac{f[x_3, x_4] - f[x_2, x_3]}{x_4 - x_2} = \frac{442 - 2}{5 - 0} = \frac{440}{5} = 88.$$

Third divided difference:-

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$$

$$= \frac{10 - 94}{2 + 4} = \frac{-84}{6} = -14$$

$$f[x_1, x_2, x_3, x_4] = \frac{f[x_2, x_3, x_4] - f[x_1, x_2, x_3]}{x_4 - x_1}$$

$$= \frac{88 - 10}{5 + 1} = \frac{78}{6} = 13$$

Fourth divided difference:-

$$f[x_0, x_1, x_2, x_3, x_4] = \frac{f[x_1, x_2, x_3, x_4] - f[x_0, x_1, x_2, x_3]}{x_4 - x_0}$$

$$= \frac{13 + 14}{5 + 4} = \frac{27}{9} = 3$$

$$\begin{aligned} \therefore P_n(x) &= 1245 + (x+4)(-404) + (x+4)(x+1)(94) + (x+4)(x+1)x(-14) \\ &\quad + (x+4)(x+1)(x)(x-2)3 \\ &= 1245 - 404(x+4) + 94(x+4)(x+1) - 14(x+4)(x+1)x \\ &\quad + 3(x+4)(x+1)x(x-2) \\ &= 1245 - 404x - 1616 + 94(x^2 + 5x + 4) - 14x(x^2 + 5x + 4) \end{aligned}$$

$$+ 3(x^2 + 5x + 4)(x^2 - 2x)$$

$$= 1245 - 404x - 1616 + 94x^2 + 470x + 376 - 14x^3 - 70x^2 - 56x$$

$$+ 3(x^4 - 2x^3 + 5x^2 - 10x^2 + 4x^2 - 8x)$$

$$= 1245 - 1616 + 376 - 404x + 470x - 56x + 94x^2 - 70x^2 - 14x^3$$

$$+ 3x^4 - 6x^3 + 15x^2 - 30x^2 + 12x^2 - 24x$$

$$\neq 5 + 42x$$

$$= 5 - 14x + 6x^2 - 5x^3 + 3x^4$$

$$= 3x^4 - 5x^3 + 6x^2 - 14x + 5$$

At $x=3$,

$$P_n(3) = 3 \cdot 3^4 - 5 \cdot 3^3 + 6 \cdot 3^2 - 14 \cdot 3 + 5$$

$$= 3 \cdot 81 - 5 \cdot 27 + 6 \cdot 9 - 42 + 5$$

$$= 243 - 135 + 54 - 42 + 5$$

$$= 302 - 177$$

$$= \underline{125}$$

12	94	106	1245
-70	100	376	1621
-30	6	-1616	-5
15	1	470	-404
-14	-6	66	-24
-6	-5	42	-56
		14	

g.

x	1	3	4	5	7	10
$f(x)$	3	31	69	131	351	1011

find $f(x)$ as a polynomial in x for the above data by Newton's divided difference formula.

Then, interpolate at $x=3.5$ and $x=8.0$

x	x_0	x_1	x_2	x_3	x_4	x_5
	1	3	4	5	7	10
$f(x)$	3	31	69	131	351	1011

$$\begin{aligned}
 P_n(x) = & f(x_0) + (x-x_0) f[x_0, x_1] + (x-x_0)(x-x_1) f[x_0, x_1, x_2] \\
 & + (x-x_0)(x-x_1)(x-x_2) f[x_0, x_1, x_2, x_3] \\
 & + (x-x_0)(x-x_1)(x-x_2)(x-x_3) f[x_0, x_1, x_2, x_3, x_4] \\
 & + (x-x_0)(x-x_1)(x-x_2)(x-x_3)(x-x_4) f[x_0, x_1, x_2, x_3, x_4, x_5]
 \end{aligned}$$

First divided difference:-

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{31 - 3}{3 - 1} = \frac{28}{2} = 14$$

$$f[x_1, x_2] = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{69 - 31}{4 - 3} = 38$$

$$f[x_2, x_3] = \frac{f(x_3) - f(x_2)}{x_3 - x_2} = \frac{131 - 69}{5 - 4} = 62$$

$$f[x_3, x_4] = \frac{f(x_4) - f(x_3)}{x_4 - x_3} = \frac{351 - 131}{7 - 5} = \frac{220}{2} = 110$$

$$f[x_4, x_5] = \frac{f(x_5) - f(x_4)}{x_5 - x_4} = \frac{1011 - 351}{10 - 7} = \frac{660}{3} = 220$$

Second divided difference:-

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{38 - 14}{4 - 1} = \frac{24}{3} = 8$$

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1} = \frac{62 - 38}{5 - 3} = \frac{24}{2} = 12$$

$$f[x_2, x_3, x_4] = \frac{f[x_3, x_4] - f[x_2, x_3]}{x_4 - x_2} = \frac{110 - 62}{7 - 4} = \frac{48}{3} = 16$$

$$f[x_3, x_4, x_5] = \frac{f[x_4, x_5] - f[x_3, x_4]}{x_5 - x_3} = \frac{220 - 110}{10 - 5} = \frac{110}{5} = 22$$

Third divided difference:-

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{12 - 8}{5 - 1} = \frac{4}{4} = 1$$

$$f[x_1, x_2, x_3, x_4] = \frac{f[x_2, x_3, x_4] - f[x_1, x_2, x_3]}{x_4 - x_1} = \frac{16 - 12}{7 - 3} = \frac{4}{4} = 1$$

$$f[x_2, x_3, x_4, x_5] = \frac{f[x_3, x_4, x_5] - f[x_2, x_3, x_4]}{x_5 - x_2} = \frac{22 - 16}{10 - 4} = \frac{6}{6} = 1$$

Fourth divided difference:-

$$f[x_0, x_1, x_2, x_3, x_4] = \frac{f[x_1, x_2, x_3, x_4] - f[x_0, x_1, x_2, x_3]}{x_4 - x_0}$$

$$= \frac{1 - 1}{7 - 1} = 0$$

$$f[x_1, x_2, x_3, x_4, x_5] = \frac{f[x_2, x_3, x_4, x_5] - f[x_1, x_2, x_3, x_4]}{x_5 - x_1}$$

$$= \frac{1 - 1}{10 - 3} = 0$$

Fifth divided difference:-

$$f[x_0, x_1, x_2, x_3, x_4, x_5] = \frac{f[x_1, x_2, x_3, x_4, x_5] - f[x_0, x_1, x_2, x_3, x_4]}{x_5 - x_0}$$

$$= 0$$

$$\therefore P_n(x) = 3 + (x-1)14 + (x-1)(x-3)8 + (x-1)(x-3)(x-4)1$$

$$= 3 + 14x - 14 + (x^2 - 3x - x + 3)8 + (x^2 - 4x + 3)(x-4)$$

$$= \underline{3} + \underline{14x} - \underline{14} + \underline{8x^2} - \underline{32x} + \underline{24} + \underline{x^3} - \underline{4x^2} - \underline{4x^2} + \underline{16x} + \underline{3x} - \underline{12}$$

$$= x^3 + x + 1$$

$$\therefore P_n(x) = x^3 + x + 1$$

$$\text{At } x = 3.5$$

$$P_n(x) = (3.5)^3 + 3.5 + 1$$

$$= 42.875 + 3.5 + 1$$

$$= 47.375$$

$$\text{At } x = 8.0$$

$$P_n(x) = 8^3 + 8 + 1$$

$$= 521$$

$\begin{array}{r} 3 + 24 \\ 27 \\ \hline -26 \\ \hline 1 \end{array}$	$\begin{array}{r} 8 \quad 14 \\ -4 \quad -32 \\ -4 \quad +16 \\ \quad +3 \\ \hline 33 \\ -32 \\ \hline 1 \end{array}$
---	---

$$\text{Ch-1: } -1.1, 1.2$$

$$\text{Ch-2: } -2.1, 2.2$$

Sampling + Quantisation.

9. Construct the Lagrange interpolation polynomial for the data.

	x_0	x_1	x_2	x_3
x	-1	1	4	7
$f(x)$	-2	0	63	342

Hence, interpolate at $x=5$

$$P_n(x) = l_0(x)f(x_0) + l_1(x)f(x_1) + l_2(x)f(x_2) + l_3(x)f(x_3)$$

We know,

$$P_n(x_i) = f(x_i)$$

$$l_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{x+1}{(-1-1)(-1-4)(-1-7)}$$

$$= \frac{(x-1)(x-4)(x-7)}{(-2)(-5)(-8)}$$

$$= -\frac{1}{80}(x^2-5x+4)(x-7)$$

$$= -\frac{1}{80}(x^3-7x^2-5x^2+35x+4x-28)$$

$$= -\frac{1}{80}(x^3-12x^2+39x-28)$$

$$l_1(x) = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} = \frac{(x+1)(x-4)(x-7)}{(1+1)(1-4)(1-7)}$$

$$= \frac{(x^2-4x+x-4)(x-7)}{2 \cdot (-3)(-6)} = \frac{(x^2-3x-4)(x-7)}{36} = \frac{x^3-7x^2-3x^2+21x-4x+28}{36}$$

$$= \frac{x^3 - 10x^2 + 17x + 28}{36}$$

$$l_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} = \frac{(x+1)(x-1)(x-7)}{(4+1)(4-1)(4-7)}$$

$$= \frac{(x^2-1)(x-7)}{5 \cdot 3 \cdot (-3)} = -\frac{x^3 - 7x^2 - x + 7}{45}$$

$$l_3(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} = \frac{(x+1)(x-1)(x-4)}{(7+1)(7-1)(7-4)}$$

$$= \frac{(x^2-1)(x-4)}{8 \cdot 6 \cdot 3} = \frac{x^3 - 4x^2 - x + 4}{144} \quad \begin{matrix} 6 \cdot 8 \\ 8 \end{matrix}$$

$$P_n(x) = l_0(x)f(x_0) + l_1(x)f(x_1) + l_2(x)f(x_2) + l_3(x)f(x_3)$$

$$= \left(-\frac{1}{80}\right)(x^3 - 12x^2 + 39x - 28)(-2) + \left(-\frac{1}{45}\right)(x^3 - 7x^2 - x + 7)63 + \frac{1}{144}(x^3 - 4x^2 - x + 4)(342)$$

$$= \left(+\frac{2}{80} - \frac{63}{45} + \frac{342}{144}\right)x^3 + \left(-\frac{24}{80} + \frac{7 \cdot 63}{45} - \frac{4 \cdot 342}{144}\right)x^2 + \left(\frac{2 \cdot 39}{80} + \frac{63}{45} - \frac{342}{144}\right)x + \left(-\frac{2 \cdot 28}{80} - \frac{7 \cdot 63}{45} + \frac{4 \cdot 342}{144}\right)$$

$$= x^3 + 0 \cdot x^2 + 0 \cdot x + (-1)$$

$$= x^3 - 1$$

$$\therefore \text{At } x=5$$

$$P_n(5) = 5^3 - 1 = 125 - 1 = 124$$

27/01/2023 Numerical Methods:-

Interpolation:

- Unevenly spaced $\begin{cases} \rightarrow \text{Lagrange} \\ \rightarrow \text{Newton's divided difference} \end{cases}$
- Evenly spaced $\begin{cases} \rightarrow \text{Lagrange} \\ \rightarrow \text{Newton's dd} \end{cases} \left. \vphantom{\begin{cases} \rightarrow \text{Lagrange} \\ \rightarrow \text{Newton's dd} \end{cases}} \right\} \text{doesn't depend on spacing}$

$$x_{i+h} = x_0 + ih$$

$$f(x_{i+h}) = f(x_0 + ih)$$

Shift operators - E .

$$E(f(x)) = f(x+h) = f(x_{i+1})$$

$$E(f(x_0)) = f(x_0+h) = f(x_1)$$

$$E(f(x_1)) = f(x_1+h) = f(x_2)$$

$$\begin{aligned}
 E^2(f(x_i)) &= E(E(f(x_i))) = E(f(x_{i+1})) = f(x_{i+2}) \\
 &= f(x_{i+1}+h) \\
 &= f(x_i+h+h) \\
 &= f(x_i+2h)
 \end{aligned}$$

$$E^2(f(x_i)) = f(x_i+2h)$$

$$E^k(f(x_i)) = f(x_i+kh)$$

$$E^{1/2}(f(x_i)) = f(x_i+\frac{1}{2}h)$$

Forward difference operator: Δ

1st
Forward
difference

$$\Delta(f(x_i)) = f(x_{i+1}) - f(x_i) = f_{i+1} - f_i$$

$$\Delta f(x_0) = f(x_1) - f(x_0) = f(x_0+h) - f(x_0) = f_1 - f_0$$

$$\Delta f(x_1) = f(x_2) - f(x_1) = f(x_1+h) - f(x_1) = f_2 - f_1$$

2nd
Forward
difference

$$\Delta^2(f(x_i)) = \Delta(\Delta(f(x_i))) = \Delta(f(x_{i+1}) - f(x_i))$$

$$= \Delta f(x_{i+1}) - \Delta f(x_i)$$

$$= f(x_{i+2}) - f(x_{i+1}) - f(x_{i+1}) + f(x_i)$$

$$= f(x_i+2h) - 2f(x_i+h) + f(x_i)$$

$$\Delta^2 f(x_1) = f(x_1+2h) - 2f(x_1+h) + f(x_1)$$

3rd
Fwd
difference

$$\Delta^3 f(x_i) = \Delta(\Delta^2 f(x_i)) = \Delta[f(x_{i+2h}) - 2f(x_{i+h}) + f(x_i)]$$

$$= \Delta f(x_{i+2h}) - 2\Delta f(x_{i+h}) + \Delta f(x_i)$$

$$= f(x_{i+2h+h}) - f(x_{i+2h}) - 2[f(x_{i+h+h}) - f(x_{i+h})] + f(x_{i+h}) - f(x_i)$$

$$= f(x_{i+3h}) - f(x_{i+2h}) - 2f(x_{i+2h}) + 2f(x_{i+h}) + f(x_{i+h}) - f(x_i)$$

$$= f(x_{i+3h}) - 3f(x_{i+2h}) + 3f(x_{i+h}) - f(x_i)$$

$$= f_{i+3} - 3f_{i+2} + 3f_{i+1} - f_i$$

Forward Difference table:-

x	$f(x)$	Δf	$\Delta^2 f$	$\Delta^3 f$	$\Delta^4 f$
x_0	$f(x_0)$				
x_1	$f(x_1)$	$\Delta f_0 = f_1 - f_0$	$\Delta^2 f_0 = \Delta f_1 - \Delta f_0$	$\Delta^3 f_0 = \Delta^2 f_1 - \Delta^2 f_0$	
x_2	$f(x_2)$	$\Delta f_1 = f_2 - f_1$	$\Delta^2 f_1 = \Delta f_2 - \Delta f_1$	$\Delta^3 f_1 = \Delta^2 f_2 - \Delta^2 f_1$	
x_3	$f(x_3)$	$\Delta f_2 = f_3 - f_2$	$\Delta^2 f_2 = \Delta f_3 - \Delta f_2$		
x_4	$f(x_4)$	$\Delta f_3 = f_4 - f_3$			

Backward difference operator:- ∇

First backward differences:-

$$\nabla f(x_i) = f(x_i) - f(x_{i-h})$$

$$\nabla f(x_1) = f(x_1) - f(x_{1-h}) = f(x_1) - f(x_0) = \Delta f(x_0)$$

$$\nabla f(x_2) = f(x_2) - f(x_2 - h) = f(x_2) - f(x_1) = \Delta f(x_1)$$

Second backward difference:-

$$\begin{aligned}\nabla^2 f(x_i) &= \nabla(\nabla f(x_i)) = \nabla [f(x_i) - f(x_i - h)] = \nabla f(x_i) - \nabla f(x_i - h) \\ &= f(x_i) - f(x_i - h) - [f(x_i - h) - f(x_i - h - h)] \\ &= f(x_i) - f(x_i - h) - f(x_i - h) + f(x_i - 2h) \\ &= f(x_i) - 2f(x_i - h) + f(x_i - 2h) \\ &= f_i - 2f_{i-1} + f_{i-2}\end{aligned}$$

3rd Backward difference:-

$$\begin{aligned}\nabla^3 f(x_i) &= \nabla(\nabla^2 f(x_i)) = \nabla [f(x_i) - 2f(x_i - h) + f(x_i - 2h)] \\ &= \nabla f(x_i) - 2\nabla f(x_i - h) + \nabla f(x_i - 2h) \\ &= f(x_i) - f(x_i - h) - 2[f(x_i - h) - f(x_i - h - h)] + f(x_i - 2h) - f(x_i - 2h - h) \\ &= f(x_i) - f(x_i - h) - 2f(x_i - h) + 2f(x_i - 2h) + f(x_i - 2h) - f(x_i - 3h) \\ &= f(x_i) - 3f(x_i - h) + 3f(x_i - 2h) - f(x_i - 3h) \\ &= f_i - 3f_{i-1} + 3f_{i-2} - f_{i-3}\end{aligned}$$

Backward difference table:-

x	$f(x)$	$\nabla f(x)$	$\nabla^2 f(x)$	$\nabla^3 f(x)$
x_0	$f(x_0)$	$\nabla f(x_1) = f(x_1) - f(x_0)$	$\nabla^2 f_2 = \nabla f_2 - \nabla f_1$	$\nabla^3 f_3 = \nabla^2 f_3 - \nabla^2 f_2$
x_1	$f(x_1)$	$\nabla f(x_2) = f(x_2) - f(x_1)$	$\nabla^2 f_3 = \nabla f_3 - \nabla f_2$	
x_2	$f(x_2)$	$\nabla f(x_3) = f(x_3) - f(x_2)$		
x_3	$f(x_3)$			

Central Difference Operators:- δ

1st Central difference:-

$$\delta(f(x_i)) = f(x_i + \frac{h}{2}) - f(x_i - \frac{h}{2}) = f_{i+1/2} - f_{i-1/2}$$

$$\delta(f(x_i + h/2)) = f(x_i + \frac{h}{2} + \frac{h}{2}) - f(x_i + \frac{h}{2} - \frac{h}{2})$$

$$= f(x_i + h) - f(x_i)$$

$$= f_{i+1} - f_i$$

$$= f_{i+1} - f_i$$

$$= \Delta f_i = \nabla f_{i+1}$$

Second central difference:-

$$\delta^2(f(x_i)) = \delta(\delta(f(x_i))) = \delta(f(x_i + h) - f(x_i)) = \delta(f(x_i + h + \frac{h}{2}) - f(x_i))$$

$$= f(x_i + h + \frac{h}{2}) - f(x_i + h - \frac{h}{2}) - [f(x_i + \frac{h}{2}) - f(x_i - \frac{h}{2})]$$

$$= f(x_i + \frac{3h}{2}) - f(x_i + \frac{h}{2}) - f(x_i + \frac{h}{2}) + f(x_i - \frac{h}{2})$$

$$= f(x_i + \frac{3h}{2}) - 2f(x_i + \frac{h}{2}) + f(x_i - \frac{h}{2}) = f_{i+3/2} - 2f_{i+1/2} + f_{i-1/2}$$

3rd central difference:

$$\delta^3 f(x_i) = \delta(\delta^2 f(x_i)) = \delta \left[f\left(x_i + \frac{3h}{2}\right) - 2f\left(x_i + \frac{h}{2}\right) + f\left(x_i - \frac{h}{2}\right) \right]$$

$$= \delta f\left(x_i + \frac{3h}{2}\right) - 2\delta f\left(x_i + \frac{h}{2}\right) + \delta f\left(x_i - \frac{h}{2}\right)$$

$$= f\left(x_i + \frac{3h}{2} + \frac{h}{2}\right) - f\left(x_i + \frac{3h}{2} - \frac{h}{2}\right) - 2 \left[f\left(x_i + \frac{h}{2} + \frac{h}{2}\right) - f\left(x_i + \frac{h}{2} - \frac{h}{2}\right) \right] + f\left(x_i - \frac{h}{2} + \frac{h}{2}\right) - f\left(x_i - \frac{h}{2} - \frac{h}{2}\right)$$

$$= f(x_i + 2h) - f(x_i + h) - 2f(x_i + h) + 2f(x_i) + f(x_i) - f(x_i - h)$$

$$= f(x_i + 2h) - 3f(x_i + h) + 3f(x_i) - f(x_i - h)$$

$$= f_{i+2} - 3f_{i+1} + 3f_i - f_{i-1}$$

Prove the following:

$$1. \delta = \nabla(1 - \nabla)^{-1/2}$$

$$2. \Delta(f_i^2) = (f_i + f_{i+1})\Delta f_i$$

$$1. \delta(f(x_i)) = f\left(x_i + \frac{h}{2}\right) - f\left(x_i - \frac{h}{2}\right)$$

$$\nabla(f(x_i)) = f(x_i) - f(x_i - h)$$

$$R.H.S = \nabla(1 - \nabla)^{-1/2}$$

$$= [f(x_i) - f(x_i - h)] [1 - f(x_i) + f(x_i - h)]^{-1/2}$$

$$= \frac{f(x_i) - f(x_i - h)}{[1 - f(x_i) + f(x_i - h)]^{1/2}}$$

$$= \frac{f(x_i) - f(x_i - h)}{[1 - f(x_i) + f(x_i - h)]^{1/2}}$$

$$\textcircled{2}. \Delta(f_i^2) = (f_i + f_{i+1}) \Delta f_i$$

$$\text{L.H.S} = \Delta(f^2(x_i)) = \Delta(f(f(x_i)))$$

$$= \Delta f^2(x_{i+1}) - f^2(x_i)$$

$$= f^2(x_{i+1}) - f^2(x_i)$$

$$= f_{i+1}^2 - f_i^2$$

$$\therefore \text{L.H.S} = \text{R.H.S} \text{ [proved]}$$

$$\text{R.H.S} = (f_i + f_{i+1}) \Delta f_i$$

$$= [f(x_i) + f(x_{i+1})] \Delta f(x_i)$$

$$= [f(x_i) + f(x_{i+1})] [f(x_{i+1}) - f(x_i)]$$

$$= f^2(x_{i+1}) - f^2(x_i)$$

$$= f_{i+1}^2 - f_i^2$$

$$f(x_i) - f(x_{i-h}) = - \left[-f(x_i) + f(x_{i-h}) + 1 - 1 \right]$$

$$= - \left[1 - f(x_i) + f(x_{i-h}) \right] + 1$$

$$\nabla(1 - \nabla)^{-1/2} = \nabla$$

$$\textcircled{1}. E = (1 - \nabla)^{-1} \Rightarrow E^{-1} = 1 - \nabla \Rightarrow \nabla = 1 - E^{-1}$$

$$\nabla f(x_i) = \nabla f(x_i) - \nabla f(x_{i-h})$$

$$= \nabla f(x_i) - \cancel{E^{-1} f(x_i)} \nabla \left(E^{-1} (f(x_i)) \right)$$

$$= \nabla f(x_i) [1 - E^{-1}]$$

$$=$$

Q.

	x_0	x_1	x_2	x_3
x	-1	0	2	3
$f(x)$	-8	3	1	12

 Obtain divided difference.

First divided difference:-

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{3 - (-8)}{0 - (-1)} = \frac{11}{1} = 11$$

$$f[x_1, x_2] = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{1 - 3}{2 - 0} = \frac{-2}{2} = -1$$

$$f[x_2, x_3] = \frac{f(x_3) - f(x_2)}{x_3 - x_2} = \frac{12 - 1}{3 - 2} = \frac{11}{1} = 11$$

Second divided difference:-

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{-1 - 11}{2 - (-1)} = \frac{-12}{3} = -4$$

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1} = \frac{11 - (-1)}{3 - 0} = \frac{12}{3} = 4$$

Third divided difference:-

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{4 - (-4)}{3 - (-1)} = \frac{8}{4} = 2$$

Newton's Divided Difference interpolation:-

$$f(x) = P_n(x) = f(x+h) - f(x) = \Delta f(x)$$

↓
Taylor series expansion

$$= [f(x) + h f'(x) + \frac{h^2}{2!} f''(x) + \dots] - f(x)$$

$$= h f'(x) + \frac{h^2}{2!} f''(x) + \dots$$

Ignoring the higher order terms we get,

$$\Delta f(x) = hf'(x).$$

$$\Delta^2 f(x) = f(x+2h) - 2f(x+h) + f(x)$$

$$= \left[\cancel{f(x)} + 2h \cancel{f'(x)} + \frac{(2h)^2}{2!} f''(x) + \dots \right] - 2 \left[\cancel{f(x)} + h \cancel{f'(x)} + \frac{h^2}{2!} f''(x) + \dots \right] + \cancel{f(x)}$$

$$= \cancel{2hf'(x)} - \cancel{hf'(x)}$$

$$= \frac{4h^2}{2!} f''(x) - \frac{2h^2}{2!} f''(x) \quad [\text{ignoring higher order terms}]$$

$$= \frac{2h^2}{2!} f''(x)$$

$$f(x) = P_n(x) = f(x_0) + (x-x_0) f[x_0, x_1] + (x-x_0)(x-x_1) f[x_0, x_1, x_2] + \dots$$

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{\Delta f(x_0)}{(x_1 - x_0)} = \frac{\Delta f_0}{x_0 + h - x_0} = \frac{\Delta f_0}{h}$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{(x_2 - x_0)} = \frac{(\Delta f_1 - \Delta f_0)}{h(x_2 - x_0)} = \frac{\Delta^2 f_0}{h(x_2 - x_0)}$$

$$= \frac{\Delta^2 f_0}{h(x_0 + 2h - x_0)} = \frac{\Delta^2 f_0}{2h^2} = \frac{\frac{2h^2}{2!} f''(x_0)}{2h^2} = \frac{f''(x_0)}{2!}$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{\frac{\Delta^2 f_1}{2h^2} - \frac{\Delta^2 f_0}{2h^2}}{3h}$$

$$= \frac{\Delta^3 f_0}{6h^3} = \frac{\Delta^3 f_0}{3! h^3}$$

$$\Delta^n f[x_0, x_1, x_2, \dots, x_n] = \frac{1}{n! h^n} \Delta^n f_0$$

$$\begin{aligned} \therefore f(x) = & f(x_0) + (x-x_0) \frac{\Delta f_0}{h} + (x-x_0)(x-x_1) \frac{\Delta^2 f_0}{2! h^2} + (x-x_0)(x-x_1)(x-x_2) \frac{\Delta^3 f_0}{3! h^3} \\ & + \dots + (x-x_0)(x-x_1) \dots (x-x_{n-1}) \frac{\Delta^n f_0}{n! h^n} \end{aligned}$$

Newton's forward difference interpolation.

Ex. 2.16, 2.17.

x	x_0	x_1	x_2	x_3	x_4	x_5
	-2	-1	0	1	2	3
$f(x)$	15	5	1	3	11	25

Construct the forward difference formula.
Hence, find $f(0.5)$.

x	$f(x)$	Δf	$\Delta^2 f$	$\Delta^3 f$
-2	15	$\Delta f_0 = -10$	$\Delta^2 f_0 = 6$	$\Delta^3 f_0 = 0$
-1	5	$\Delta f_1 = -4$	$\Delta^2 f_1 = 6$	$\Delta^3 f_1 = 0$
0	1	$\Delta f_2 = 2$	$\Delta^2 f_2 = 6$	$\Delta^3 f_2 = 0$
1	3	$\Delta f_3 = 8$	$\Delta^2 f_3 = 6$	$\Delta^3 f_3 = 0$
2	11	$\Delta f_4 = 14$		
3	25			

$$h = -1 - (-2) = 1$$

$\therefore$ Forward difference formula is,

$$\begin{aligned}
 f(x) &= 15 + (x+2)\frac{(-10)}{1} + (x+2)(x+1)\frac{6}{2!1^2} + (x+2)(x+1)(x-0)\frac{0}{3!1^3} \\
 &= 15 - 10x - 20 + (x^2 + x + 2x + 2)\frac{6}{2} + 0 \\
 &= 15 - 10x - 20 + 3x^2 + 9x + 6 \\
 &= 3x^2 - x + 1
 \end{aligned}$$

$$\therefore f(0.5) = 3(0.5)^2 - (0.5) + 1 = 1.25$$

Newton's Backward difference interpolation:-

• Newton's Divided difference polynomial:-

$$\begin{aligned}
 P_n(x) = f(x) &= f(x_0) + (x-x_0)f[x_0, x_1] + (x-x_0)(x-x_1)f[x_0, x_1, x_2] + \\
 &\quad + (x-x_0)(x-x_1)\dots(x-x_{n-1})f[x_0, x_1, x_2, \dots, x_n]
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow f(x) &= f(x_n) + (x-x_n)f[x_n, x_{n-1}] + (x-x_n)(x-x_{n-1})f[x_n, x_{n-1}, x_{n-2}] \\
 &\quad + \dots + (x-x_n)(x-x_{n-1})\dots(x-x_0)f[x_n, x_{n-1}, \dots, x_0]
 \end{aligned}$$

For the data,

x	$f(x)$	1st d.d	2nd d.d	3rd d.d
x_0	$f(x_0)$	$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$	$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$	$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$
x_1	$f(x_1)$	$f[x_1, x_2] = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$	$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}$	$= \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$
x_2	$f(x_2)$	$f[x_2, x_3] = \frac{f(x_3) - f(x_2)}{x_3 - x_2}$		$\div (x_3 - x_0)$
x_3	$f(x_3)$			

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{f(x_0) - f(x_1)}{x_0 - x_1} = f[x_1, x_0]$$

$$f[x_n, x_{n-1}] = f[x_{n-1}, x_n] = \frac{1}{h} \nabla f(x_n) \quad [\text{from Taylor series expansion}]$$

$$f[x_n, x_{n-1}, x_{n-2}] = f[x_{n-2}, x_{n-1}, x_n] = \frac{1}{2! h^2} \nabla^2 f(x_n)$$

and neglect neglecting higher order terms.

$$\therefore f(x) = f(x_n) + (x - x_n) f[x_n, x_{n-1}] + (x - x_n)(x - x_{n-1}) f[x_n, x_{n-1}, x_{n-2}]$$

$$+ \dots + (x - x_n)(x - x_{n-1}) \dots (x - x_1) f[x_n, x_{n-1}, \dots, x_0]$$

$$= f_n + (x - x_n) \frac{\nabla f_n}{h} + (x - x_n)(x - x_{n-1}) \frac{\nabla^2 f_n}{2! h^2} + \dots$$

$$+ (x - x_n)(x - x_{n-1}) \dots (x - x_1) \frac{\nabla^n f_n}{n! h^n}$$

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
0.1	1.4			
0.2	1.56	$\Delta f_0 = 0.16$	$\Delta^2 f_0 = 0.04$	$\Delta^3 f_0 = 0$
0.3	1.76	$\Delta f_1 = 0.2$	$\Delta^2 f_1 = 0.04$	$\Delta^3 f_1 = 0$
0.4	2.0	$\Delta f_2 = 0.24$	$\Delta^2 f_2 = 0.04$	
0.5	2.28	$\Delta f_3 = 0.28$		

$$\begin{array}{r} 2.00 \\ 0.76 \\ \hline 0.24 \end{array}$$

$$h = 0.2 - 0.1 = 0.1$$

∴ forward difference formula is,

$$f(x) = f(x_0) + (x-x_0) \frac{\Delta f(x_0)}{h} + (x-x_0)(x-x_1) \frac{\Delta^2 f(x_0)}{2! h^2} + \dots + (x-x_0)(x-x_1) \dots (x-x_{n-1}) \frac{\Delta^n f(x_0)}{n! h^n}$$

$$= 1.4 + (x-0.1) \frac{0.16}{0.1} + (x-0.1)(x-0.2) \frac{0.04}{2!(0.1)^2} + (x-0.1)(x-0.2)(x-0.3) 0$$

$$= 1.4 + (x-0.1) 1.6 + (x^2 - 0.2x - 0.1x + 0.02) \frac{0.04 \times 2}{2 \times 0.01} + 0$$

$$= 1.4 + 1.6x - 0.16 + 2x^2 - 0.6x + 0.04$$

$$= \cancel{2x^2} + \cancel{2.2x} + \cancel{1.6} \text{ (Ans)} = 2x^2 + 1x + 1.28 \text{ (Ans)}$$

x	$f(x)$	$\nabla f(x)$	$\nabla^2 f(x)$	$\nabla^3 f(x)$
x_0 0.1	1.4			
x_1 0.2	1.56	$\nabla f(x_1) = 0.16$	$\nabla^2 f_2 = 0.04$	$\nabla^3 f_3 = 0$
x_2 0.3	1.76	$\nabla f(x_2) = 0.2$	$\nabla^2 f_3 = 0.04$	$\nabla^3 f_4 = 0$
x_3 0.4	2.0	$\nabla f(x_3) = 0.24$	$\nabla^2 f_4 = 0.04$	
x_4 0.5	2.28	$\nabla f(x_4) = 0.28$		

$$h = 0.2 - 0.1 = 0.1$$

∴ Backward difference formula is,

$$f(x) = f_n + (x-x_n) \frac{\nabla f_n}{h} + (x-x_n)(x-x_{n-1}) \frac{\nabla^2 f_n}{2! h^2} + \dots + (x-x_n)(x-x_{n-1}) \dots (x-x_1) \frac{\nabla^n f_n}{n! h^n}$$

$$= 2.28 + (x-0.5) \frac{0.28}{0.1} + (x-0.5)(x-0.4) \frac{0.04}{2!(0.1)^2} + 0$$

$$= 2.28 + (x-0.5) 2.8 + (x^2 - 0.4x - 0.5x + 0.2) \frac{0.04 \times 2}{2 \times 0.01}$$

$$= 2.28 + 2.8x - 1.4 + 2x^2 - 1.8x + 0.4 = 2x^2 + 1x + 1.28 \text{ (Ans)}$$

Lagrange Polynomials-

$$\begin{aligned}l_0(x) &= \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)} \\&= \frac{(x-0.2)(x-0.3)(x-0.4)(x-0.5)}{(0.1-0.2)(0.1-0.3)(0.1-0.4)(0.1-0.5)} \\&= \frac{(x-0.2)(x-0.3)(x-0.4)(x-0.5)}{(-0.1)(-0.2)(-0.3)(-0.4)} \\&= \frac{(x-0.2)(x-0.3)(x-0.4)(x-0.5)}{0.0024}\end{aligned}$$

$$\begin{aligned}l_1(x) &= \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)} \\&= \frac{(x-0.1)(x-0.3)(x-0.4)(x-0.5)}{(0.2-0.1)(0.2-0.3)(0.2-0.4)(0.2-0.5)} \\&= \frac{(x-0.1)(x-0.3)(x-0.4)(x-0.5)}{0.1(-0.1)(-0.2)(-0.3)} \\&= \frac{(x-0.1)(x-0.3)(x-0.4)(x-0.5)}{-0.0006}\end{aligned}$$

$$\begin{aligned}l_2(x) &= \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)} \\&= \frac{(x-0.1)(x-0.2)(x-0.4)(x-0.5)}{(0.2)(0.1)(-0.1)(-0.2)} \\&= \frac{(x-0.1)(x-0.2)(x-0.4)(x-0.5)}{0.0004}\end{aligned}$$

$$\begin{aligned}l_3(x) &= \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)} \\&= \frac{(x-0.1)(x-0.2)(x-0.3)(x-0.5)}{(0.3)(0.2)(0.1)(-0.1)} \\&= \frac{(x-0.1)(x-0.2)(x-0.3)(x-0.5)}{-0.0006}\end{aligned}$$

$$\begin{aligned}l_4(x) &= \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)} \\&= \frac{(x-0.1)(x-0.2)(x-0.3)(x-0.4)}{(0.4)(0.3)(0.2)(0.1)} \\&= \frac{(x-0.1)(x-0.2)(x-0.3)(x-0.4)}{0.0024}\end{aligned}$$

$$P_n(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) + L_3(x)f(x_3) + L_4(x)f(x_4)$$

$$= (1.4) \frac{(x-0.2)(x-0.3)(x-0.4)(x-0.5)}{0.0024} + (1.56) \frac{(x-0.1)(x-0.3)(x-0.4)(x-0.5)}{(-0.0006)}$$

$$+ (1.76) \frac{(x-0.1)(x-0.2)(x-0.4)(x-0.5)}{0.0004} + (2.0) \frac{(x-0.1)(x-0.2)(x-0.3)(x-0.5)}{(-0.0006)}$$

$$+ (2.28) \frac{(x-0.1)(x-0.2)(x-0.3)(x-0.4)}{0.0024}$$

$$= \frac{(x-0.2)(x-0.3)(x-0.4)}{0.0024} \left[(1.4)(x-0.5) + (2.28)(x-0.1) \right]$$

$$+ \frac{(x-0.1)(x-0.3)(x-0.5)}{(-0.0006)} \left[(1.56)(x-0.4) + (2.0)(x-0.2) \right]$$

$$+ (1.76) \frac{(x-0.1)(x-0.2)(x-0.4)(x-0.5)}{0.0004}$$

$$= \frac{(x^2 - 0.5x + 0.06)(3.68x^2 - 2.4x + 0.3712)}{0.0024}$$

$$- \frac{(x^2 - 0.4x + 0.03)(3.56x^2 - 2.804x + 0.512)}{0.0006}$$

$$+ (1.76) \frac{(x^2 - 0.3x + 0.02)(x^2 - 0.9x + 0.2)}{0.0004}$$

$$(1.4x - 0.7 + 2.28x - 0.228)$$

$$= (3.68x - 0.928)(x-0.4)$$

$$(1.56x - 0.624 + 2.0x - 0.4)$$

$$= (3.56x - 1.024)(x-0.5)$$

$$= 3.68x^2 - 1.472x - 0.928x + 0.3712$$

$$= (3.68x^2 - 2.4x + 0.3712)$$

$$= 3.56x^2 - 1.78x - 1.024x + 0.512$$

$$= 3.56x^2 - 2.804x + 0.512$$

Spline Interpolation:-

Spline- Let, a given interval $[a, b]$ be sub-divided into n -sub-intervals $[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$ such that,

$$a = x_0 < x_1 < x_2 \dots < x_n = b$$

$x_0, x_1, \dots, x_n$ are called nodes/knots.

$x_1, x_2, \dots, x_{n-1}$ are called internal nodes.

Spline function:-

Spline function of degree n , with nodes $x_0, x_1, \dots, x_n$ is a function $F(x)$ satisfying the following properties:

(i) $F(x_i) = f(x_i)$; $i = 0, 1, \dots, n$ (Interpolation condition).

(ii) On each sub-interval $[x_{i-1}, x_i]$, $1 \leq i \leq n$, $F(x)$ is a polynomial of degree n , $n=3$ for cubic spline.

(iii) $F(x)$ and its first $(n-1)$ derivatives are continuous on $(a, b) \Rightarrow F(x), F'(x), F''(x)$ are continuous.

Cubic Spline:-

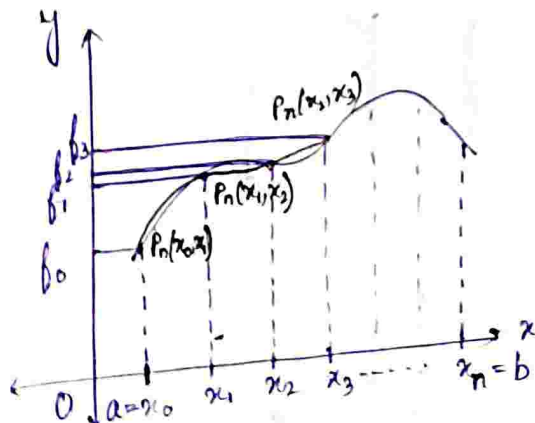
$$F(x) = P_i(x) = a_i x^3 + b_i x^2 + c_i x + d_i$$

$$F'(x) = 3a_i x^2 + 2b_i x + c_i$$

$$F''(x) = 6a_i x + 2b_i$$

All continuous.

Continuity of $f(x)$ & $F'(x)$



SPLINE:-

03/02/2023

① $F(x_i) = f(x_i) \quad \forall i = 0, 1, 2, \dots, n$

② On each interval $[x_{i-1}, x_i]$, $F(x)$ is a polynomial of degree n .
(Cubic $\Rightarrow n=3$)

③ $F(x)$ and $F'(x), \dots, F^{(n-1)}(x)$ are all continuous on (a, b) .

$$F(x) = P_i(x) = a_i x^3 + b_i x^2 + c_i x + d_i \quad \text{on } [x_{i-1}, x_i]$$

four unknowns: a_i, b_i, c_i, d_i .

④ Continuity of $F(x)$ on (a, b)

on $[x_{i-1}, x_i]$, $P_i(x_i) = f(x_i) = a_i x_i^3 + b_i x_i^2 + c_i x_i + d_i \quad \forall i = 1, 2, \dots, n-1$

on $[x_i, x_{i+1}]$, $P_{i+1}(x_i) = f(x_i) = a_{i+1} x_i^3 + b_{i+1} x_i^2 + c_{i+1} x_i + d_{i+1}$

⑤ Continuity of $F'(x)$:

on $[x_{i-1}, x_i] \Rightarrow F'(x_i) = 3a_i x_i^2 + 2b_i x_i + c_i$

on $[x_i, x_{i+1}] \Rightarrow F'(x_i) = 3a_{i+1} x_i^2 + 2b_{i+1} x_i + c_{i+1}$

$$3a_i x_i^2 + 2b_i x_i + c_i = 3a_{i+1} x_i^2 + 2b_{i+1} x_i + c_{i+1} \quad \forall i = 1, 2, \dots, n-1$$

⑥ Continuity of $F''(x)$:

$$6a_i x_i + 2b_i = 6a_{i+1} x_i + 2b_{i+1} \quad \forall i = 1, 2, \dots, n-1$$

$F''(x)$ is a linear function.

Consider the interval $[x_{i-1}, x_i]$ and use Lagrange's interpolation,

$$F''(x) = \frac{x - x_i}{x_{i-1} - x_i} F''(x_{i-1}) + \frac{x - x_{i-1}}{x_i - x_{i-1}} F''(x_i)$$

$$= \frac{x_i^0 - x}{x_i^0 - x_{i-1}^0} F''(x_{i-1}^0) + \frac{x - x_{i-1}^0}{x_i^0 - x_{i-1}^0} F''(x_i^0)$$

$$\int \frac{(a-x)dx}{(b-c)} = \int \frac{adx}{b-c} - \int \frac{x}{b-c} dx$$

$$= \frac{x_i^0 - x}{x_i^0 - x_{i-1}^0} F''(x_{i-1}^0) + \frac{x - x_{i-1}^0}{x_i^0 - x_{i-1}^0} F''(x_i^0) \xrightarrow{\textcircled{i}} = \frac{x_i^0 - x}{x_i^0 - x_{i-1}^0} M_{i-1} + \frac{x - x_{i-1}^0}{x_i^0 - x_{i-1}^0} M_i \xrightarrow{\textcircled{i}} = \frac{ax}{b-c} + \frac{x^2}{2(b-c)}$$

where $F''(x_{i-1}^0) = M_{i-1}$ and $F''(x_i^0) = M_i$

Integrating we get,

$$F'(x) = \frac{(x_i^0 - x)^2}{2(x_i^0 - x_{i-1}^0)} M_{i-1} + \frac{(x - x_{i-1}^0)^2}{2(x_i^0 - x_{i-1}^0)} M_i + a, \text{ where } a \text{ is integrating constant.}$$

$\xrightarrow{\textcircled{ii}}$

Again integrating we get,

$$F(x) = \frac{(x_i^0 - x)^3}{6(x_i^0 - x_{i-1}^0)} M_{i-1} + \frac{(x - x_{i-1}^0)^3}{6(x_i^0 - x_{i-1}^0)} M_i + ax + b, \text{ where } a \text{ and } b \text{ both are constants.}$$

$\xrightarrow{\textcircled{iii}}$

$$F(x_{i-1}^0) = f(x_{i-1}^0) \text{ and } F(x_i^0) = f(x_i^0)$$

denote $x_i^0 - x_{i-1}^0 = h_i$

$$ax + b = c(x_i^0 - x) + d(x - x_{i-1}^0), \text{ where } a = d - c, b = cx_i^0 - dx_{i-1}^0$$

$$F(x_{i-1}^0) = f(x_{i-1}^0) = f_{i-1}^0$$

$$F(x) = \frac{(x_i^0 - x)^3}{6h_i^0} M_{i-1} + \frac{(x - x_{i-1}^0)^3}{6h_i^0} M_i + c(x_i^0 - x) + d(x - x_{i-1}^0)$$

$$F(x_{i-1}^0) = \frac{(x_i^0 - x_{i-1}^0)^3}{6h_i^0} M_{i-1} + c(x_i^0 - x_{i-1}^0)$$

$$= \frac{h_i^{0^3}}{6h_i^0} M_{i-1} + ch_i^0$$

$$= \frac{h_i^{0^2}}{6} M_{i-1} + ch_i^0$$

$$\therefore ch_i^0 = F(x_{i-1}^0) - \frac{h_i^{0^2}}{6} M_{i-1}$$

$$\Rightarrow c = \frac{1}{h_i^0} \left[F(x_{i-1}^0) - \frac{h_i^{0^2}}{6} M_{i-1} \right]$$

$$F(x_i) = f(x_i) = f_i$$

$$\Rightarrow F(x) = \frac{(x_i - x_{i-1})^3}{6h_i} M_i + d(x_i - x_{i-1})$$

$$\Rightarrow d = \frac{1}{h_i} \left[f_i - \frac{h_i^2}{6} M_i \right]$$

$$[x_{i-1}, x_i] \quad \therefore F_i(x) = \frac{1}{6h_i} \left[(x_i - x)^3 M_{i-1} + (x - x_{i-1})^3 M_i \right] + \frac{(x_i - x)}{h_i} \left[f_{i-1} - \frac{h_{i-1}^2}{6} M_{i-1} \right] \quad \text{--- (A)}$$

$$+ \frac{(x - x_{i-1})}{h_i} \left[f_i - \frac{h_i^2}{6} M_i \right] \quad \text{where } M_{i-1} = F''(x_{i-1})$$

$$M_i = F''(x_i)$$

$$[x_i, x_{i+1}]$$

$$F_{i+1}(x) = \frac{1}{6h_{i+1}} \left[(x_{i+1} - x)^3 M_i + (x - x_i)^3 M_{i+1} \right] + \frac{(x_{i+1} - x)}{h_{i+1}} \left[f_i - \frac{h_i^2}{6} M_i \right]$$

$$+ \frac{(x - x_i)}{h_{i+1}} \left[f_{i+1} - \frac{h_{i+1}^2}{6} M_{i+1} \right] \quad \text{--- (B)}$$

$$h_{i+1} = x_{i+1} - x_i$$

Differentiating (A) and (B), $F'(x)$ is continuous,

$$\lim_{\epsilon \rightarrow 0} F'_i(x_i - \epsilon) = \lim_{\epsilon \rightarrow 0} F'_{i+1}(x_i + \epsilon)$$

$$F'_i(x) = \frac{1}{6h_i} \left[-3(x_i - x)^2 M_{i-1} + 3(x - x_{i-1})^2 M_i \right] + \frac{(-1)}{h_i} \left[f_{i-1} - \frac{h_{i-1}^2}{6} M_{i-1} \right]$$

$$+ \frac{1}{h_i} \left[f_i - \frac{h_i^2}{6} M_i \right]$$

$$F'_{i+1}(x) = \frac{1}{6h_i} \left[3(x_i - x_{i-1})^2 M_{i+1} \right] - \frac{1}{h_{i+1}} \left[f_{i-1} - \frac{h_{i-1}^2}{6} M_{i-1} \right] + \frac{1}{h_i} \left[f_i - \frac{h_i^2}{6} M_i \right]$$

$$= \frac{1}{2h_i} h_i^2 M_{i+1} - \frac{1}{h_{i+1}} \left[f_{i-1} - \frac{h_{i-1}^2}{6} M_{i-1} \right] + \frac{1}{h_i} \left[f_i - \frac{h_i^2}{6} M_i \right]$$

$$= \frac{h_i}{2} M_{i+1} - \frac{1}{h_{i+1}} f_{i-1} + \frac{h_{i-1}}{6} M_{i-1} + \frac{f_i}{h_i} - \frac{h_i}{6} M_i$$

$$f'_{i+1}(\alpha) =$$

$$\frac{h_i}{2} M_{i+1} - \frac{1}{h_i} f_{i-1} + \frac{h_i}{6} M_{i-1} + \frac{f_i}{h_i} - \frac{h_i}{6} M_i = -\frac{h_{i+1}}{2} M_i - \frac{f_i}{h_{i+1}} + \frac{h_{i+1}}{6} M_i + \frac{1}{h_{i+1}} f_{i+1} - \frac{h_{i+1}}{6} M_{i+1}$$

Equispaced data, $h_i = h_{i+1} = h$, $x_i = x_0 + ih$

$$\therefore \frac{h}{2} M_i - \frac{f_{i-1}}{h} + \frac{h}{6} M_{i-1} + \frac{f_i}{h} - \frac{h}{6} M_i + \frac{h M_i}{2} + \frac{f_i}{h} - \frac{h}{6} M_i + \frac{f_{i+1}}{h} + \frac{h}{6} M_{i+1} = 0$$

$$\Rightarrow M_i \left[\frac{h}{2} - \frac{h}{6} + \frac{h}{2} - \frac{h}{6} \right] + M_{i-1} \left[\frac{h}{6} \right] + \left[\frac{h}{6} \right] M_{i+1} + f_i \left(\frac{1}{h} + \frac{1}{h} \right)$$

$$- \frac{f_{i-1}}{h} + \frac{f_{i+1}}{h} = 0$$

$$\frac{2h}{2} - \frac{2h}{6}$$

$$\Rightarrow \frac{2h}{3} M_i + \frac{h}{6} M_{i-1} + \frac{h}{6} M_{i+1} + \frac{2}{h} f_i$$

$$= \frac{3h-h}{3}$$

$$- \frac{f_{i-1}}{h} + \frac{f_{i+1}}{h} = 0$$

$$= \frac{2h}{3}$$

$$\Rightarrow M_{i-1} + 4M_i + M_{i+1} = \frac{6}{h^2} (f_{i+1} - 2f_i + f_{i-1}), \quad i=1, 2, \dots, n-1$$

Example:

x	0	1	2
$f(x)$	-1	3	29

$$M_{i-1} = M_0, \quad M_i = M_1, \quad M_{i+1} = M_2$$

$$M_0 = 0, \quad M_2 = 0 \quad \text{Cubic spline.}$$

$$M_0 + 4M_1 + M_2 = \frac{6}{h^2} (f_2 - 2f_1 + f_0), \quad i=1, 2, \dots, n-1$$

$$\therefore M_0 = 0, M_2 = 0$$

$$\Rightarrow 4M_1 = 6(29 - 2 \times 3 - 1)$$

$$\Rightarrow 4M_1 = 6(29 - 6 - 1)$$

$$\Rightarrow M_1 = \frac{6^3}{4} \times \frac{22}{2} = 33$$

$$F_1(x) = \frac{1}{6} [(x_1 - x)^3 \cdot 0 + (x - x_0)^3 M_1] + \frac{(x_1 - x)}{1} \left[f_0 - \frac{1}{6} M_0 \right] + \frac{(x - x_0)}{1} \left[f_1 - \frac{1}{6} M_1 \right]$$

$$= \frac{1}{6} (x-0)^3 \cdot 33 + \frac{(1-x)}{1} [(-1) - 0] + \frac{(x-0)}{1} \left[3 - \frac{1}{6} \cdot 33 \right]$$

$$= \frac{11}{2} x^3 - 1 + x + 3x - \frac{11}{2} x$$

$$= \frac{11}{2} x^3 + (1 + 3 - \frac{11}{2}) x - 1$$

$$= \frac{11}{2} x^3 + \frac{(2 + 6 - 11)}{2} x - 1$$

$$= \frac{11}{2} x^3 + \left(-\frac{3}{2}\right) x - 1$$

$$= \frac{11}{2} x^3 - \frac{3}{2} x - 1 \quad \text{for interval } [0, 1]$$

$$F_2(x) = \frac{1}{6} [(x_2 - x)^3 M_1 + (x - x_1)^3 M_2] + \frac{(x_2 - x)}{1} \left[f_1 - \frac{1}{6} M_1 \right] + \frac{(x - x_1)}{1} \left[f_2 - \frac{1}{6} M_2 \right]$$

$$= \frac{1}{6} [(2-x)^3 \cdot 33] + (2-x) \left[3 - \frac{1}{6} \cdot 33 \right] + (x-1) \left[29 - \frac{1}{6} \cdot 0 \right]$$

$$= \frac{1}{6} (2-x)^3 \cdot 33 - (2-x) \left[+3 - \frac{11}{2} \right] + (x-1) \cdot 29$$

We have, $M_{i-1} + 4M_i + M_{i+1} = \frac{6}{h^2} (f_{i+1} - 2f_i + f_{i-1})$, $i=1, 2, \dots, n-1$

For, $i=1$,

$$M_0 + 4M_1 + M_2 = \frac{6}{h^2} (f_2 - 2f_1 + f_0)$$

$$\because M_0 = 0, M_2 = 0$$

$$\Rightarrow 4M_1 = \frac{6}{h^2} (29 - 2 \times 3 - 1)$$

$$\Rightarrow 4M_1 = 6(29 - 6 - 1)$$

$$\Rightarrow M_1 = \frac{6^3 \times 22}{4^2} = 33$$

$$F_1(x) = \frac{1}{6} [(x_1 - x)^3 \cdot 0 + (x - x_0)^3 M_1] + \frac{(x_1 - x)}{1!} [f_0 - \frac{1}{6} M_0]$$

$$+ \frac{(x - x_0)}{1} [f_1 - \frac{1}{6} M_1]$$

$$= \frac{1}{6} (x - 0)^3 \cdot 33 + \frac{(1 - x)}{1} [(-1) - 0] + \frac{(x - 0)}{1} [3 - \frac{1}{6} \cdot 33]$$

$$= \frac{11}{2} x^3 - 1 + x + 3x - \frac{11}{2} x$$

$$= \frac{11}{2} x^3 + (1 + 3x - \frac{11}{2}) x - 1$$

$$= \frac{11}{2} x^3 + \frac{(2 + 6 - 11)}{2} x - 1$$

$$= \frac{11}{2} x^3 + \left(-\frac{3}{2}\right) x - 1$$

$$= \frac{11}{2} x^3 - \frac{3}{2} x - 1 \quad \text{for interval } [0, 1].$$

$$F_2(x) = \frac{1}{6} [(x_2 - x)^3 M_1 + (x - x_1)^3 M_2] + \frac{(x_2 - x)}{1!} [f_1 - \frac{1}{6} M_1] + \frac{(x - x_1)}{1} [f_2 - \frac{1}{6} M_2]$$

$$= \frac{1}{6} [(2 - x)^3 \cdot 33] + (2 - x) [3 - \frac{1}{6} \cdot 33] + (x - 1) [29 - \frac{1}{6} \cdot 33]$$

$$= \frac{1}{6} (2 - x)^3 \cdot 33 - (2 - x) [3 - \frac{11}{2}] + (x - 1) \cdot 29$$

Numerical Differentiation:-

Interpolation $\begin{cases} \text{evenly spaced points} \\ \text{unevenly spaced points} \end{cases}$

Evenly spaced points:-

Derivatives based on Newton's forward Difference formula.

$x_i = x_0 + ih$, h : step length.

$(x_i, f(x_i))$, $\forall i = 0, 1, \dots, n$

$$f(x) = f(x_0) + (x-x_0) \frac{\Delta f_0}{1!h} + (x-x_0)(x-x_1) \frac{\Delta^2 f_0}{2!h^2} + \dots \\ \dots + (x-x_0)(x-x_1) \dots (x-x_{n-1}) \frac{\Delta^n f_0}{n!h^n}$$

If $x = x_0 + sh \Rightarrow (x-x_0) = sh$

Also, $x = x_0 + sh$

$x_1 = x_0 + 1h$

for, $x = x_k$, $\frac{(x_k - x_0)}{h} = s$.

$$\Rightarrow x - x_1 = (s-1)h$$

$$\text{Then, } f(x) = f(x_0) + s \frac{\Delta f_0}{1!} + s(s-1) \frac{\Delta^2 f_0}{2!} + \dots + s(s-1)(s-2) \dots (s-n+1) \frac{\Delta^n f_0}{n!}$$

$$\therefore \frac{df}{dx} = \frac{df}{ds} \cdot \frac{ds}{dx} = \frac{1}{h} \cdot \frac{df}{ds} ; \frac{d}{dx} \left(\frac{df}{dx} \right) = \frac{d}{dx} \left(\frac{1}{h} \cdot \frac{df}{ds} \right) = \frac{1}{h^2} \frac{d}{ds} \left(\frac{df}{ds} \right)$$

$$s = \frac{(x-x_0)}{h}$$

$$\frac{ds}{dx} = \frac{1}{h}$$

$$= \frac{1}{h} \frac{d}{dx} \left(\frac{df}{ds} \right) \\ = \frac{1}{h} \frac{d}{ds} \cdot \left(\frac{df}{ds} \right) \left(\frac{ds}{dx} \right)$$

$$\therefore \frac{df(x)}{dx} = \frac{1}{h} \frac{df}{ds} = \frac{1}{h} \left[0 + \frac{\Delta f_0}{1!} + \frac{\Delta^2 f_0}{2!} (2s-1) + \dots + \frac{\Delta^n f_0}{n!} (\dots) \right]$$

$$\frac{d^2 f(x)}{dx^2} = \frac{1}{h^2} \frac{d}{ds} \left(\frac{df}{ds} \right) = \frac{1}{h^2} \left[0 + \Delta^2 f_0 + \frac{1}{6} (6s-6) \Delta^3 f_0 + \dots \right]$$

$$f'(x_0) = \frac{1}{h} \left[\Delta f_0 + \left(-\frac{\Delta^2 f_0}{2!} \right) \right] = \frac{1}{h} \left[\Delta f_0 - \frac{\Delta^2 f_0}{2} + \frac{1}{3} \Delta^3 f_0 - \frac{1}{4} \Delta^4 f_0 + \frac{1}{5} \Delta^5 f_0 - \dots \right]$$

Ignoring higher order terms we have,

$$f'(x_0) = \frac{1}{h} \Delta f_0 = \frac{1}{h} [f(x_1) - f(x_0)]$$

$$f'(x_k) = \frac{1}{h} \Delta f_k = \frac{1}{h} [f(x_{k+1}) - f(x_k)]$$

$$f''(x_k) = \frac{1}{h^2} [f(x_{k+2}) - 2f(x_{k+1}) + f(x_k)]$$

$$f''(x_0) = \frac{1}{h^2} \Delta^2 f_0$$

$$= \frac{1}{h^2} [\Delta f_1 - \Delta f_0]$$

$$= \frac{1}{h^2} [f(x_2) - f(x_1) - f(x_1) + f(x_0)]$$

$$= \frac{1}{h^2} [f(x_2) - 2f(x_1) + f(x_0)]$$

$$f''(x_0) = \frac{1}{h^2} [\Delta^2 f_0 - \Delta^3 f_0 + \frac{11}{2} \Delta^4 f_0 - \frac{5}{6} \Delta^5 f_0 + \dots]$$

$$f'(x_0) = \frac{1}{h} [\Delta f_0 - \frac{\Delta^2 f_0}{2}]$$

$$= \frac{1}{h} [f_1 - f_0 - \frac{1}{2} (f_2 - 2f_1 + f_0)]$$

$$= \frac{1}{h} [f - f_0 - \frac{f_2}{2} + f_1 - \frac{f_0}{2}]$$

$$= \frac{1}{h} [2f_1 - \frac{3}{2} f_0 - \frac{f_2}{2}]$$

$$\therefore f'(x_0) = \frac{1}{2h} [4f_1 - 3f_0 - f_2]$$

$$f'(x_k) = \frac{1}{2h} [-3f(x_k) + 4f(x_{k+1}) - f(x_{k+2})]$$

$$f''(x_0) = \frac{1}{h^2} \Delta^2 f_0$$

Q. x	f(x)	Δf	$\Delta^2 f$	$\Delta^3 f$
1	1	7		
2	8	19	12	
3	27	37	18	6
4	64			

$$\frac{df}{dx} = f'(x) = \frac{1}{h} [\Delta f_0 - \frac{1}{2} \Delta^2 f_0 + \frac{1}{3} \Delta^3 f_0]$$

$$= 1 \left[7 - \frac{1}{2} \cdot 12 + \frac{1}{3} \cdot 6 \right]$$

$$= 1 (1 + 2) = 3$$

$$x=1 = x_0$$

$$h=1$$

$$x = x_0 + sh$$

Find $\frac{df(x)}{dx}$ at $x=1$ using
Newton's Forward difference
method.

Numerical Differentiations:- (Evenly spaced points)

06/02/2023 .

1. Forward differences

2. Backward differences:- $(x_i, f(x_i)) \forall i=0, 1, 2, \dots, n$

$$x_i = x_0 + ih$$

$$f(x) = f(x_n) + (x-x_n) \frac{1}{1!h} \nabla f(x_n) + (x-x_n)(x-x_{n-1}) \frac{\nabla^2 f(x_n)}{2!h^2} + \dots$$

$$\dots + (x-x_n) \dots (x-x_1) \frac{1}{n!h^n} \nabla^n f(x_n)$$

Let, x be any point near x_n .

$$x - x_n = sh \Rightarrow x = x_n + sh \quad \text{--- (i)}$$

$$\therefore f(x) = f(x_n) + \frac{s}{1!} \nabla f(x_n) + \frac{s(s+1)}{2!} \nabla^2 f(x_n) + \dots + \frac{s(s+1)(s+2) \dots (s+n-1)}{n!} \nabla^n f(x_n)$$

$$x_{n-1} = x_0 + (n-1)h \quad \text{--- (ii)}$$

$$x_n = x_0 + nh \Rightarrow x_0 = x_n - nh \quad \text{--- (iii)}$$

$$x_n - x_{n-1} = h(n - (n-1))$$

$$x - x_{n-1} = x - (x_0 + (n-1)h) \quad \text{[From (ii)]}$$

$$= x_n + sh - x_0 - (n-1)h \quad \text{[From (i)]}$$

$$= x_n + sh - x_n + nh - nh + h \quad \text{[From (iii)]}$$

$$\text{From (i), } s = \frac{(x - x_n)}{h} < 0$$

Hence, higher terms would tend to 0.

$$\Rightarrow \frac{ds}{dx} = \frac{1}{h}$$

First derivative:- $= h(s+1)$

$$\frac{df}{dx} = \frac{df}{ds} \cdot \frac{ds}{dx} = \frac{1}{h} \frac{df}{ds}$$

$$\therefore f'(x) = \frac{1}{h} \left[\cancel{f(x_n)} + \frac{\nabla f(x_n)}{1!} + \frac{\nabla^2 f(x_n)}{2!} (2s+1) + \frac{(3s^2+6s+2)}{3!} \nabla^3 f(x_n) + \dots \right]$$

Approx to the $f'(x)$ at $x = x_n \rightarrow s = 0$

$$f'(x_n) = \frac{1}{h} \nabla f(x_n) + \frac{1}{2} \nabla^2 f(x_n) + \frac{1}{3} \nabla^3 f(x_n) + \dots$$

$$x - x_{n-1} \Rightarrow (s+1)h = 0 \Rightarrow s+1 = 0 \Rightarrow s = -1$$

$$f'(x_{n-1}) = \frac{1}{h} \left[\nabla f(x_n) + \frac{1}{2} (-2+1) \nabla^2 f(x_n) - \frac{1}{6} \nabla^3 f(x_n) - \dots \right]$$

Second derivative:-

$$\frac{d^2 f}{dx^2} = \frac{d}{dx} \left(\frac{df}{dx} \right) \rightarrow \frac{1}{h} \frac{d}{ds} \left(\frac{df}{ds} \right) \frac{ds}{dx} \neq \frac{d}{dx}$$

$$= \frac{d}{dx} \left(\frac{1}{h} \frac{df}{ds} \right) = \frac{1}{h} \frac{d}{ds} \left(\frac{df}{ds} \right) \frac{ds}{dx} = \frac{1}{h^2} \frac{d}{ds} \left(\frac{df}{ds} \right)$$

$$= \frac{1}{h^2} \left[\nabla^2 f(x_n) + \frac{1}{6} (6s+6) \nabla^3 f(x_n) + \dots \right]$$

$$= \frac{1}{h^2} \left[\nabla^2 f(x_n) + \nabla^3 f(x_n) + \frac{11}{12} \nabla^4 f(x_n) + \frac{5}{6} \nabla^5 f(x_n) + \dots \right] \text{ if } s=0$$

If $x = x_n \Rightarrow s = 0$

$$f''(x_n) = \frac{1}{h^2} \left[\nabla^2 f(x_n) + \nabla^3 f(x_n) + \dots \right]$$

If $x = x_{n-1} \Rightarrow s = -1$

$$f''(x_n) = \frac{1}{h^2} \left[\nabla^2 f(x_n) - \frac{1}{12} \nabla^4 f(x_n) - \frac{1}{12} \nabla^5 f(x_n) + \dots \right]$$

Error:- $|f'(x) - \tilde{f}'(x)|$ where $\tilde{f}'(x)$ = computing from Newton's forward/backward difference

$f'(x)$ = actual derivative of $f(x)$.

Q.

x	$f(x)$	$\nabla f(x)$	$\nabla^2 f(x)$	$\nabla^3 f(x)$
1.0	3.7183			
1.5	5.4817	1.7634		
2.0	8.9891	2.9074	1.144	
2.5	13.1825	4.7934	1.886	

 $h = 0.5$

$$f(x) = 13.1825 + \frac{s}{1!} 4.7934 + \frac{3(s+1)}{2!} 1.886$$

$$f'(x) = \frac{1}{h} \left[4.7934 + (2s+1) \frac{1.886}{2!} \right]$$

$$= \frac{1}{0.5} \left[4.7934 + (2s+1) \frac{1.886}{2!} \right]$$

Find $f'(2.5)$, $f'(2)$ and $f''(2.5)$ using Newton's backward difference method.

Original $f(x) \Rightarrow f(x) = e^{x+1}$

Find error in each case.

$$x - x_n = sh$$

For, $x = 2.5$

$$2.5 - 2.5 = s(0.5)$$

$$\Rightarrow s = 0$$

$f'(2.5)$

$$= \frac{1}{0.5} \left[4.7934 + \frac{1.886}{2!} \right]$$

$$= 11.4728$$

For, $x = 2$

$x - x_n = sh$

$\Rightarrow 2 - 2.5 = s(0.5)$

$\Rightarrow -0.5 = s(0.5)$

$\Rightarrow s = -1$

~~$f'(x)$~~ for ~~$x_1 = 2$~~

$\therefore f(x) = f(x_n) + \frac{s}{1!} \nabla f(x_n) + \frac{s(s+1)}{2!} \nabla^2 f(x_n)$

$+ \frac{s(s+1)(s+2)}{3!} \nabla^3 f(x_n) + \dots$

$f'(x) = \left[\frac{f(x_n)}{1!} + \frac{\nabla f(x_n)}{2!} (2s+1) + \frac{(3s^2+6s+2)}{3!} \nabla^2 f(x_n) \right] \frac{1}{h}$

$f''(x) = \frac{1}{h^2} \left[\nabla^2 f(x_n) \cdot 2 + \frac{\nabla^3 f(x_n)}{3!} (6s+6) + \dots \right]$

$\therefore f'(2.5) = \frac{1}{0.5} \left[\frac{4.7934}{1!} + \frac{1.886}{2!} \cdot 1 + \frac{2}{3 \times 2} 0.742 \right]$

$= \frac{1}{0.5} [4.7934 + 0.943 + 0.2473]$

$= 11.9674$

$f''(2.5) = \frac{1}{(0.5)^2} \left[\frac{1.886}{2!} \cdot 2 + \frac{0.742}{3 \times 2} \cdot 6 \right]$

$= 10.512$

$f'(2) = \frac{1}{0.5} \left[\frac{4.7934}{1!} + \frac{1.886}{2!} (2+1) + \frac{(3-6+2)}{3!} 0.742 \right]$

$= \frac{1}{0.5} [4.7934 - 0.943 - 0.1236]$

$= 7.4536$

$f(x) = e^x + 1 \quad f''(x) = e^x \quad \therefore \text{Errors:}$

$f'(x) = e^x$

Case-1 $|f'(x) - \tilde{f}'(x)| = |12.182494 - 11.9674|$
 $= 0.215094$

$f'(2.5) = 12.182494$

$f'(2) = 7.3890561$

Case-2 $|f'(x) - \tilde{f}'(x)| = |7.3890561 - 7.4536|$
 $= 0.0645439$

$f''(2.5) = 12.182494$

Case-3) $|f''(x) - \tilde{f}''(x)| = |12.182494 - 10.512| = 1.670494$.

3. Divided differences:-

$(x_i, f(x_i))$, $i = 0, 1, 2, \dots, n$

$$f(x) = f(x_0) + (x-x_0)f[x_0, x_1] + (x-x_0)(x-x_1)f[x_0, x_1, x_2] + \dots + (x-x_0)(x-x_1)\dots(x-x_{n-1})f[x_0, x_1, x_2, \dots, x_n]$$

$$f'(x) = f[x_0, x_1] + (x-x_0) + (x-x_1) f[x_0, x_1, x_2] + (x-x_1)(x-x_2) + (x-x_0)(x-x_2) f[x_0, x_1, x_2, x_3] + \dots$$

$$f''(x) = 2f[x_0, x_1, x_2] + 2[(x-x_0)+(x-x_1)+(x-x_2)]f[x_0, x_1, x_2, x_3] + \dots$$

NUMERICAL INTEGRATION:-

Integration of a function $f(x)$,

$$] = \int_a^b f(x) dx \quad \left[\text{Assuming both } f(x) \text{ and } w(x)f(x) \text{ are integrable in the interval } [a, b] \right]$$

$$J = \int_a^b w(x) f(x) dx \rightarrow \text{weighted integration.}$$

$\downarrow$
 $w(x) \geq 0$, weighted function.
 in (a, b)

$$I = \int_a^b w(x) f(x) dx = \sum_{k=0}^n \lambda_k f(x_k) = \lambda_0 f(x_0) + \lambda_1 f(x_1) + \dots + \lambda_n f(x_n) \quad \text{--- (1)}$$

Order of a method:- An integration method of the form (1) is said to be of order p , if it produces exact result for all polynomials of degree less than or equal to p .

$$\Rightarrow R_n(x^m) = \int_a^b w(x) x^m dx - \sum_{k=0}^n \lambda_k x_k^m = 0$$

$$f(x) = 1, x, x^2, \dots, x^p$$

Error term is obtained for $f(x) = x^{p+1}$

Error constant: $c = \int_a^b w(x) x^{p+1} dx - \sum_{k=0}^n \lambda_k x_k^{p+1}$

$$R_n(f) = \int_a^b w(x) f(x) dx - \sum_{k=0}^n \lambda_k f(x_k)$$

$$= \frac{c}{(p+1)!} f^{(p+1)}(\xi), \quad a < \xi < b.$$

Bound on the error will be $|R_n(f)| \leq \frac{|c|}{(p+1)!} \max_{a < x < b} |f^{(p+1)}(x)|$.

Integration Rules based on uniform Mesh spacing:-

Assume, $w(x) = 1$ and x_k 's are equispaced such that,
 $x_0 = a$, $x_n = b$.

$$\therefore h = \frac{b-a}{n}$$

Then, $\boxed{I = \int_a^b w(x) f(x) dx = \sum_{k=0}^n \lambda_k f(x_k)}$ Quadrature Formula.

$$\Rightarrow \int_a^b f(x) dx = \sum_{k=0}^n \lambda_k f(x_k)$$

- x_k 's are called abscissas.
- $f(x_k)$'s are called ordinates.
- λ_k 's are weights of the integration rule of our quadrature formula.

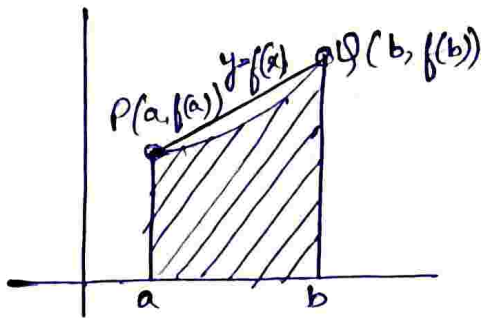
This is Newton-Cotes Integration rules.

λ_k 's are called Cotes number.

⊕

Numerical Integration:-

① Uniform Mesh Spacing - (Evenly spaced points).



Trapezium Rule:-

Use Newton's forward difference formula;

$$f(x) = f(x_0) + \frac{1}{h}(x-x_0)\Delta f(x_0) \quad \text{where } x_0=a, x_1=b$$

$$h=b-a$$

$$I = \int_a^b f(x) dx$$

$$= \int_a^b \left[f(x_0) + \frac{1}{h}(x-x_0)\Delta f(x_0) \right] dx$$

$$= f(x_0) \int_a^b dx + \frac{\Delta f(x_0)}{h} \int_a^b (x-x_0) dx$$

$$= f(x_0) [x]_{x_0}^{x_1} + \frac{\Delta f(x_0)}{h} \left[\frac{x^2}{2} - x_0 x \right]_{x_0}^{x_1}$$

$$= f(x_0)(x_1-x_0) + \frac{\Delta f(x_0)}{h} \left[\frac{x_1^2}{2} - x_0 x_1 - \frac{x_0^2}{2} + x_0^2 \right]$$

$$= f(x_0)(x_1-x_0) + \frac{\Delta f(x_0)}{2h} [x_1^2 - 2x_0 x_1 - x_0^2 + 2x_0^2]$$

$$= f(x_0)(x_1-x_0) + \frac{\Delta f(x_0)}{2h} (x_1^2 - 2x_0 x_1 + x_0^2)$$

$$= f(x_0)(x_1-x_0) + \frac{\Delta f(x_0)}{2h} (x_1-x_0)^2$$

$$= f(x_0)(x_1-x_0) + \frac{\Delta f(x_0)}{2(x_1-x_0)} (x_1-x_0)^2 = f(x_0)(x_1-x_0) + \frac{\Delta f(x_0)}{2} (x_1-x_0)$$

$$= (x_1-x_0) \left[f(x_0) + \frac{\Delta f(x_0)}{2} \right]$$

$$= (x_1 - x_0) \left(f_0 + \frac{1}{2} \Delta f_0 \right)$$

$$= (x_1 - x_0) \left[f_0 + \frac{1}{2} (f_1 - f_0) \right]$$

$$= (x_1 - x_0) \left[f_0 + \frac{1}{2} f_1 - \frac{1}{2} f_0 \right]$$

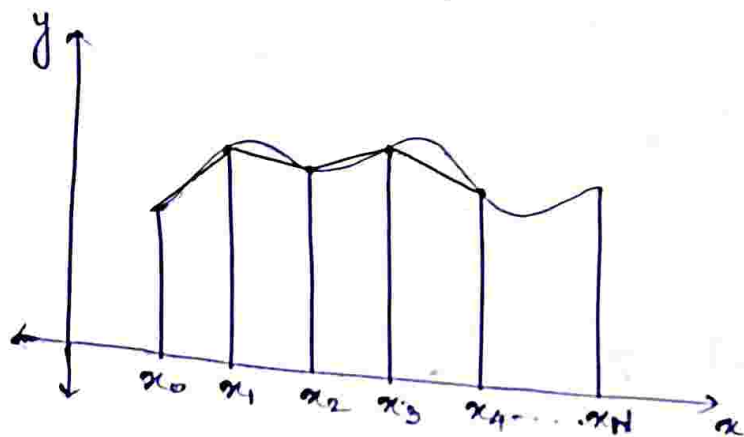
$$= (x_1 - x_0) \frac{1}{2} (f_1 + f_0)$$

$$= \frac{b-a}{2} [f(b) + f(a)]$$

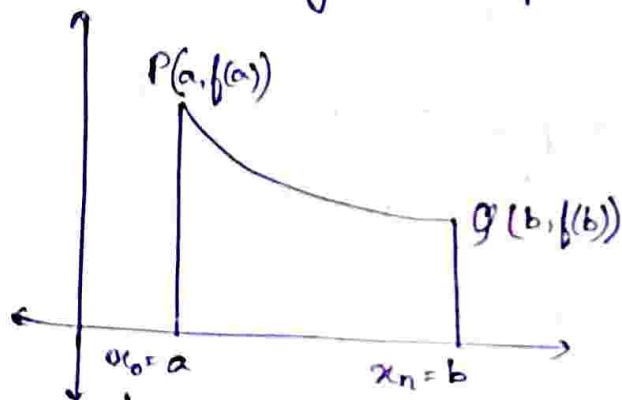
$$\therefore I = \int_a^b f(x) dx = \frac{b-a}{2} [f(b) + f(a)]$$

= area under the trapezium trapezoid with width $b-a$ and $f(b)$ & $f(a)$ as ordinates.

~ area under the curve defined by $y=f(x)$ with ordinates $x=a$ and $x=b$.



Using Lagrange's interpolation:-



$$f(x) = l_0(x) f(x_0) + l_1(x) f(x_1)$$

$$= \frac{(x-x_1)}{(x_0-x_1)} f(x_0) + \frac{(x-x_0)}{(x_1-x_0)} f(x_1)$$

$$= \frac{(x-b)}{(a-b)} f(x_0) + \frac{(a-x)}{(a-b)} f(x_1)$$

$$= \frac{1}{(a-b)} [x f(a) - b f(a) + a f(b) - x f(b)]$$

$$= \frac{1}{a-b} [x f(a) - b f(a) + a f(b) - x f(b)]$$

$$= \frac{1}{a-b} \left[\int_a^b [x (f(a) - f(b)) + a f(b) - b f(a)] dx \right] = \frac{1}{a-b} [x (f(a) - f(b)) + a f(b) - b f(a)]$$

$$= \frac{1}{a-b} \left[[f(a) - f(b)] \frac{x^2}{2} + a f(b) x - b f(a) x \right]_a^b$$

$$= \frac{1}{a-b} \left[[f(a) - f(b)] \frac{b^2}{2} + ab f(b) - b^2 f(a) - \left[[f(a) - f(b)] \frac{a^2}{2} + a^2 f(b) - ab f(a) \right] \right]$$

$$\begin{aligned}
&= \frac{1}{a-b} \left[[f(a) - f(b)] \left(\frac{b^2}{2} - \frac{a^2}{2} \right) + \cancel{ab} f(b) - b^2 f(a) - a^2 f(b) \right] \\
&\quad + ab f(a) \\
&= \frac{1}{\cancel{(a-b)}} [f(a) - f(b)] \frac{\cancel{(b-a)}(b+a)}{2} + \frac{1}{\cancel{(a-b)}} [b f(a) (\cancel{a-b}) + a f(b) (\cancel{b-a})] \\
&= [f(b) - f(a)] \frac{(b+a)}{2} + b f(a) - a f(b) \\
&= \frac{1}{2} \left[[f(b) - f(a)](b+a) + 2b f(a) - 2a f(b) \right] \\
&= \frac{1}{2} \left[b f(b) + a f(b) - b f(a) - a f(a) + 2b f(a) - 2a f(b) \right] \\
&= \frac{1}{2} \left[b f(b) + \cancel{a f(b)} - a f(b) + b f(a) - \cancel{a f(a)} \right] \\
&= \frac{1}{2} \left[b (f(b) + f(a)) - a (f(b) + f(a)) \right] \\
&= \frac{1}{2} [f(b) + f(a)](b-a) \\
&= \frac{1}{2}(b-a) [f(b) + f(a)] \quad , \text{ Trapezoidal Rule.}
\end{aligned}$$

Q. Find the approx. value of J , using trapezoidal rule with 2, 4, 8 equal sub-ordinates for given $J = \int_0^1 \frac{dx}{1+x}$.

When, $N=2$, $h = \frac{b-a}{N} = \frac{1-0}{2} = \frac{1}{2}$, Nodes $(0, \frac{1}{2}, 1)$

When, $N=4$, $h = \frac{b-a}{N} = \frac{1-0}{4} = \frac{1}{4}$, Nodes $(0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, 1)$

When, $N=8$, $h = \frac{b-a}{N} = \frac{1-0}{8} = \frac{1}{8}$, Nodes $(0, \frac{1}{8}, \frac{2}{8}, \frac{3}{8}, \dots, \frac{7}{8}, 1)$

For, $N=2$, for $(0, \frac{1}{2})$

$$\begin{aligned}
\int_0^{\frac{1}{2}} f(x) dx &= \frac{\frac{1}{2} - 0}{2} \left(\frac{2}{3} + 1 \right) \\
&= \frac{1}{4} \cdot \frac{2+3}{3} = \frac{5}{12}
\end{aligned}$$

$$\begin{aligned}
f(x) &= \frac{1}{1+x} & f(0) &= 1 \\
f\left(\frac{1}{2}\right) &= \frac{1}{1+\frac{1}{2}} = \frac{2}{3} \\
f(1) &= \frac{1}{2}
\end{aligned}$$

for $(\frac{1}{2}, 1)$

$$\int_{\frac{1}{2}}^1 f(x) dx = \frac{1 - \frac{1}{2}}{2} \left(\frac{1}{2} + \frac{2}{3} \right) = \frac{1}{4} \cdot \frac{(3+4)}{6} = \frac{7}{24}$$

$$\therefore \int_0^1 f(x) dx = \int_0^{\frac{1}{2}} f(x) dx + \int_{\frac{1}{2}}^1 f(x) dx = \frac{5}{12} + \frac{7}{24} = \frac{10+7}{24} = \frac{17}{24} = 0.70833$$

For, $N=4$, we have,

interval $(0, \frac{1}{4})$, $(\frac{1}{4}, \frac{2}{4})$, $(\frac{2}{4}, \frac{3}{4})$, $(\frac{3}{4}, 1)$

So, $f(0) = 1$

$$f(\frac{3}{4}) = \frac{1}{1 + \frac{3}{4}} = \frac{1}{\frac{7}{4}} = \frac{4}{7}$$

$$f(\frac{1}{4}) = \frac{1}{1 + \frac{1}{4}} = \frac{1}{\frac{5}{4}} = \frac{4}{5}$$

$$f(1) = \frac{1}{1+1} = \frac{1}{2}$$

$$f(\frac{2}{4}) = \frac{1}{1 + \frac{2}{4}} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$$

$$\therefore \int_0^{\frac{1}{4}} f(x) dx = \frac{\frac{1}{4} - 0}{2} (4 + 1) = \frac{1}{8} \cdot \frac{(4+5)}{5} = \frac{9}{40}$$

$$\int_{\frac{1}{4}}^{\frac{2}{4}} f(x) dx = \frac{\frac{2}{4} - \frac{1}{4}}{2} \left(\frac{2}{3} + \frac{4}{5} \right) = \frac{1}{8} \cdot \frac{(10+12)}{15} = \frac{22}{120}$$

$$\int_{\frac{2}{4}}^{\frac{3}{4}} f(x) dx = \frac{\frac{3}{4} - \frac{2}{4}}{2} \left(\frac{4}{7} + \frac{2}{3} \right) = \frac{1}{8} \cdot \frac{(12+14)}{21} = \frac{26}{168}$$

$$\int_{\frac{3}{4}}^1 f(x) dx = \frac{1 - \frac{3}{4}}{2} \left(\frac{1}{2} + \frac{4}{7} \right) = \frac{1}{8} \cdot \frac{(7+8)}{14} = \frac{15}{112}$$

$$\therefore \int_0^1 f(x) dx = \int_0^{\frac{1}{4}} f(x) dx + \int_{\frac{1}{4}}^{\frac{2}{4}} f(x) dx + \int_{\frac{2}{4}}^{\frac{3}{4}} f(x) dx + \int_{\frac{3}{4}}^1 f(x) dx$$

$$= \frac{9}{40} + \frac{22}{120} + \frac{26}{168} + \frac{15}{112}$$

$$= 0.69702$$

8	40, 120, 168, 112
5	5, 19, 21, 14
7	1, 3, 21, 14
3	1, 3, 3, 2
	1, 1, 1, 2

For, $N=8$, intervals are,

$$(0, \frac{1}{8}), (\frac{1}{8}, \frac{2}{8}), (\frac{2}{8}, \frac{3}{8}), (\frac{3}{8}, \frac{4}{8}), (\frac{4}{8}, \frac{5}{8}), (\frac{5}{8}, \frac{6}{8}), (\frac{6}{8}, \frac{7}{8}), (\frac{7}{8}, 1)$$

$$f(0) = 1 \qquad f(\frac{4}{8}) = \frac{1}{1+\frac{4}{8}} = \frac{8}{12} \qquad f(\frac{7}{8}) = \frac{1}{1+\frac{7}{8}} = \frac{8}{15}$$

$$f(\frac{1}{8}) = \frac{1}{1+\frac{1}{8}} = \frac{8}{9}$$

$$f(\frac{5}{8}) = \frac{1}{1+\frac{5}{8}} = \frac{8}{13}$$

$$f(1) = \frac{1}{1+1} = \frac{1}{2}$$

$$f(\frac{2}{8}) = \frac{1}{1+\frac{2}{8}} = \frac{8}{10}$$

$$f(\frac{6}{8}) = \frac{1}{1+\frac{6}{8}} = \frac{8}{14}$$

$$f(\frac{3}{8}) = \frac{1}{1+\frac{3}{8}} = \frac{8}{11}$$

$$\therefore \int_0^{\frac{1}{8}} f(x) dx = \frac{\frac{1}{8} - 0}{2} \left(\frac{8}{9} + 1 \right) = \frac{1}{16} \cdot \frac{17}{9} = \frac{17}{144}$$

$$\int_{\frac{1}{8}}^{\frac{2}{8}} f(x) dx = \frac{\frac{2}{8} - \frac{1}{8}}{2} \left(\frac{8}{10} + \frac{8}{9} \right) = \frac{1}{16} \cdot \frac{72+80}{90} = \frac{152}{1440}$$

$$\int_{\frac{2}{8}}^{\frac{3}{8}} f(x) dx = \frac{\frac{3}{8} - \frac{2}{8}}{2} \left(\frac{8}{11} + \frac{8}{10} \right) = \frac{1}{16} \cdot \frac{80+88}{110} = \frac{168}{1760}$$

$$\int_{\frac{3}{8}}^{\frac{4}{8}} f(x) dx = \frac{\frac{4}{8} - \frac{3}{8}}{2} \left(\frac{8}{12} + \frac{8}{11} \right) = \frac{1}{16} \cdot \frac{88+96}{132} = \frac{184}{2112}$$

$$\int_{\frac{4}{8}}^{\frac{5}{8}} f(x) dx = \frac{1}{16} \cdot \left(\frac{8}{13} + \frac{8}{12} \right) = \frac{1}{16} \cdot \frac{96+104}{156} = \frac{200}{2496}$$

$$\int_{\frac{5}{8}}^{\frac{6}{8}} f(x) dx = \frac{1}{16} \left(\frac{8}{14} + \frac{8}{13} \right) = \frac{1}{16} \cdot \frac{104+112}{182} = \frac{216}{2912}$$

$$\int_{\frac{6}{8}}^{\frac{7}{8}} f(x) dx = \frac{1}{16} \left(\frac{8}{15} + \frac{8}{14} \right) = \frac{1}{16} \cdot \frac{112+120}{210} = \frac{232}{3360}$$

$$\int_{\frac{7}{8}}^1 f(x) dx = \frac{1}{16} \left(\frac{1}{2} + \frac{8}{15} \right) = \frac{1}{16} \cdot \frac{15+16}{30} = \frac{31}{480}$$

$$\therefore \int_0^1 f(x) dx = \frac{17}{144} + \frac{152}{1440} + \frac{168}{1760} + \frac{184}{2112} + \frac{200}{2196} + \frac{216}{2912} + \frac{232}{3360} + \frac{31}{480}$$

$$= 0.40618 + 0.28793$$

$$= 0.69411$$

$$\int_0^1 f(x) dx = \int_0^1 \frac{dx}{1+x} = [\log(1+x)]_0^1 = \log e^2 - \log e^1 = 0.693147$$

Error in $N=2$,

$$|0.693147 - 0.70833| = 0.01518$$

Error in $N=4$,

$$|0.693147 - 0.69702| = 0.00387$$

Error in $N=8$,

$$|0.69314 - 0.69411| = 0.00096$$

■ Simpson's $\frac{1}{3}$ rd Rule:-

$$[a, b], -I = \int_a^b f(x) dx$$

$$h = \frac{b-a}{2}, x_0 = a, x_1 = \frac{a+b}{2}, x_2 = b$$

$$\int_a^b f(x) dx = \int_{x_0}^{x_2} f(x) dx = \int_{x_0}^{x_2} \left[f_0 + \frac{1}{h}(x-x_0) \Delta f_0 + \frac{1}{2h^2}(x-x_0)(x-x_1) \Delta^2 f_0 \right] dx$$

[By using forward difference formula]

$$= f_0 \int_{x_0}^{x_2} dx + \frac{\Delta f_0}{h} \int_{x_0}^{x_2} (x-x_0) dx + \frac{\Delta^2 f_0}{2h^2} \int_{x_0}^{x_2} (x-x_0)(x-x_1) dx$$

$$= f_0 [x]_{x_0}^{x_2} + \frac{\Delta f_0}{h} \left[\frac{x^2}{2} - x x_0 \right]_{x_0}^{x_2} + \frac{\Delta^2 f_0}{2h^2} \left[\frac{x^3}{3} - x_1 \frac{x^2}{2} - x_0 \frac{x^2}{2} + x x_0 x_1 \right]_{x_0}^{x_2}$$

$$= f_0 (x_2 - x_0) + \frac{\Delta f_0}{h} \left[\frac{x_2^2}{2} - x_2 x_0 - \frac{x_0^2}{2} + x_0^2 \right] + \frac{\Delta^2 f_0}{2h^2} \left[\frac{x_2^3}{3} - x_1 \frac{x_2^2}{2} - x_0 \frac{x_2^2}{2} + x_0 x_1 x_2 - \frac{x_0^3}{3} + x_1 \frac{x_0^2}{2} + x_0 \frac{x_0^2}{2} - x_0^2 x_1 \right]$$

$$\begin{aligned}
&= f_0(x_2 - x_0) + \frac{\Delta f_0}{2h} (x_2^2 - 2x_0x_2 + x_0^2) \\
&\quad + \frac{\Delta^2 f_0}{2h^2} \left[\frac{x_2^3 - x_0^3}{3} - \frac{(x_0 + x_1)(x_2^2 - x_0^2)}{2} + (x_0x_1(x_2 - x_0)) \right] \\
&= f_0(x_2 - x_0) + \frac{(f_1 - f_0)}{2h} (x_2 - x_0)^2 + \frac{\Delta^2 f_0}{2h^2} (x_2 - x_0) \left[\frac{x_2^2 + x_2x_0 + x_0^2}{3} \right. \\
&\quad \left. - \frac{(x_0 + x_1)(x_2 + x_0)}{2} + x_0x_1 \right] [2f_1 - f_0] \\
&= f_0(x_2 - x_0) + \frac{(f_1 - f_0)}{2 \cdot h} (2h)^2 + \frac{1}{2h^2} \cdot 2h
\end{aligned}$$

$$h = \frac{x_2 - x_0}{2}$$

$$\Rightarrow 2h = x_2 - x_0$$

$$\int_a^b f(x) dx = \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Initial Value Problems for ODE:-

$$y' = f(x, y), \quad y(x_0) = b_0$$

$$y'' = f(x, y, y'), \quad y(x_0) = b_0, \quad y'(x_0) = b_1$$

Solve:-

Reduce the 2nd order equation to 1st order system.

2nd order equation:-

$$a_0(x)y''(x) + a_1(x)y'(x) + a_2(x)y(x) = r(x).$$

$$y(x_0) = b_0, \quad y'(x_0) = b_1$$

Assume, $y = u_1$, then $u_1' = y' = u_2$

$$a_0(x)u_2' + a_1(x)u_2 + a_2(x)u_1 = r(x)$$

$$\Rightarrow u_2' = \frac{r(x) - a_1(x)u_2 - a_2(x)u_1}{a_0(x)}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}' = \begin{bmatrix} f(x, y_1, y_2) \\ f(x, y_1, y_2) \end{bmatrix},$$

$$\begin{bmatrix} y_1(x_0) \\ y_2(x_0) \end{bmatrix} = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

Reduce:- $x^2y'' + (2x+1)y' + 3y = 6$, $y(1) = 2$, $y'(1) = 0.5$

$$u_1 = y, \quad u_2 = y'$$

$$x^2u_2' + (2x+1)u_2 + 3u_1 = 6$$

$$\Rightarrow u_2' = \frac{1}{x^2} [6 - 3u_1 - (2x+1)u_2]$$

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}' = \begin{bmatrix} u_2 \\ \frac{1}{x^2} [6 - 3u_1 - (2x+1)u_2] \end{bmatrix}; \quad \begin{bmatrix} u_1(x_0) \\ u_2(x_0) \end{bmatrix} = \begin{bmatrix} 2 \\ 0.5 \end{bmatrix}$$

① Single step method: ^{Solution at} x_{i+1} using solution at x_i (increment-1)

② Multi-step method: -

▪ Taylor series Methods for IVP:-

Expand $y(x)$ in Taylor series about any point x_i , with Lagrange form of remainder.

$$y(x) = y(x_i) + (x-x_i)y'(x_i) + \frac{1}{2!}(x-x_i)^2 y''(x_i) + \dots + \frac{1}{p!}(x-x_i)^p y^{(p)}(x_i) + \frac{1}{(p+1)!}(x-x_i)^{p+1} y^{(p+1)}(x_i + \theta h)$$

where $0 < \theta < 1$, $x \in [x_0, b]$
 b is the point upto which the solution is required.

y_i = denote the numerical solution.

$y_p(x)$ = exact solution

$$[x_i, x_{i+1}], h = x_{i+1} - x_i$$

Exact solution

$$y(x_{i+1}) = y(x_i) + h y'(x_i) + \frac{h^2}{2!} y''(x_i) + \dots + \frac{h^p}{p!} y^{(p)}(x_i) + \frac{h^{p+1}}{(p+1)!} y^{(p+1)}(x_i + \theta h)$$

we can neglect this error term as it is much higher ordered.

▪ Explicit Methods:-

Approximate solution

$$y_{i+1} = y_i + h y'_i + \frac{h^2}{2!} y''_i + \dots + \frac{h^p}{p!} y^{(p)}_i + \frac{h^{p+1}}{(p+1)!} y^{(p+1)}_i$$

Truncation errors:- $\frac{h^{p+1}}{(p+1)!} y^{(p+1)}(x_i + \theta h)$

If $p=1$, $y_{i+1} = y_i + h y'_i$ [Euler's Method]

Truncation errors:- (for Euler's method)

$$\begin{aligned} & \frac{h^2}{2!} y''(x_i + \theta h) \\ &= \frac{h^2}{2!} y''(x_i) + \frac{h^3}{3!} y'''(x_i) + \dots \end{aligned}$$

Q. $yy' = x$, $y(0) = 1$; $0 \leq x \leq 0.8$, $h = 0.2$.

Euler's Method:-

$$y_{i+1} = y_i + h y_i'$$

$$\Rightarrow y_{i+1} = y_i + h \cdot \frac{x_i}{y_i}$$

$$y_i y_i' = x_i$$

$$\Rightarrow y_i' = \frac{x_i}{y_i}$$

Exact solution:-

$$yy' = x$$

~~$$y' = \frac{x}{y}$$~~

$$\Rightarrow y \cdot \frac{dy}{dx} = x$$

$$\Rightarrow y dy = x dx$$

Integrating both sides,

$$\frac{y^2}{2} = \frac{x^2}{2} + \frac{C}{h}$$

~~$$\Rightarrow y^2 = x^2 + C$$~~

$$\Rightarrow \text{At } x=0, y=1$$

$$\therefore \frac{1}{2} = 0 + C$$

$$\Rightarrow C = \frac{1}{2}$$

$$\therefore \frac{y^2}{2} = \frac{x^2}{2} + \frac{1}{2}$$

$$\Rightarrow y = \sqrt{x^2 + 1}$$

$$x_4 = 0.8 \quad y_4 = y(0.8) = \sqrt{(0.8)^2 + 1}$$

$$= \sqrt{1.64} = 1.28062$$

$$\therefore \text{Error} = |1.28062 - 1.22436|$$

$$= 0.05626$$

$$x_0 = 0, h = 0.2$$

$$y_0 = 1$$

$$x_1 = 0 + 0.2 = 0.2$$

$$y_0 = 1$$

~~$$x_2 = 0.2 + 0.2 = 0.4$$~~

$$y_1 =$$

$$x_2 = 0 + 2 \times 0.2 = 0.4$$

$$x_3 = 0 + 3 \times 0.2 = 0.6$$

$$x_4 = 0 + 4 \times 0.2 = 0.8$$

$$y_0 = 1$$

$$y_1 = y_0 + h \cdot \frac{x_0}{y_0}$$

$$= 1 + 0.2 \cdot \frac{0}{1} = 1$$
~~$$= 1.2$$~~

$$y_2 = y_1 + h \cdot \frac{x_1}{y_1}$$

$$= 1 + 0.2 \cdot \frac{(0.2)}{1}$$

$$= 1 + 0.04$$

$$= 1.04$$

$$y_3 = y_2 + h \cdot \frac{x_2}{y_2}$$

$$= 1.04 + 0.2 \cdot \frac{(0.4)}{1.04}$$

$$= 1.04 + \frac{0.08}{1.04}$$

$$= 1.1692$$

$$y_4 = y_3 + h \cdot \frac{x_3}{y_3}$$

$$= 1.1692 + (0.2) \cdot \frac{(0.6)}{1.1692}$$

$$= 1.22436$$

Q. $y' = x(y+1)$, $y(0)=1$, Taylor series method of order 2.
 $h=0.1$ compute $y(0.2)$.

$$\hookrightarrow y_{i+1} = y_i + h y'_i + \frac{h^2}{2!} y''_i$$

$$y' = \frac{dy}{dx}$$

$$y'' = \frac{d}{dx}(xy+x) = xy' + y+1$$

$$\frac{d}{dx}(xy+x) = y + xy'$$

$$y'' = \frac{d}{dx}(xy+x) = xy' + y+1, \quad x_0=0, y_0=1$$

$$y_0' = 0$$

$$x_0 = 0$$

$$y_0'' = 0 + 1 + 1 = 2$$

$$x_1 = 0 + 0.1 = 0.1$$

$$y_1' = x_1 y_1 + x_1 = 0.1(1) + 0.1 = 0.201$$

$$y_1'' = x_1 y_1' + y_1 + 1 = (0.1)(0.201) + (1.01) + 1 = 2.0301$$

$$y_2 = y_1$$

$$y_1 = y_0 + h y_0' + \frac{h^2}{2!} y_0'' = 1 + (0.1)0 + \frac{(0.1)^2}{2!} 2 = 1 + 0.01 = 1.01$$

$$y' = x(y+1)$$

$$\Rightarrow \frac{dy}{dx} = x(y+1)$$

$$\Rightarrow \int \frac{dy}{y+1} = \int x dx$$

$$\Rightarrow \log_e(y+1) = \frac{x^2}{2} + C$$

$$y_2 = y_1 + h y_1' + \frac{h^2}{2!} y_1''$$

$$= 1.01 + (0.1)(0.201) + \frac{(0.1)^2}{2!}(2.0301) \text{ for } x=0, y=0$$

$$\ln(y+1) = 0 + C$$

$$= 1.0402505$$

$$\Rightarrow \ln 1 = C$$

$$\Rightarrow C = 0$$

$$\therefore \ln(y+1) = \frac{x^2}{2}$$

$$x_2 = 0.2$$

$$y(0.2) = e^{\frac{(0.2)^2}{2}} - 1$$

$$= 0.020139$$

$$\Rightarrow y+1 = e^{\frac{x^2}{2}}$$

$$\Rightarrow y = e^{\frac{x^2}{2}} - 1$$

$$\text{Error} = |1.0402505 - 0.020139|$$

$$= 1.0201115$$

$$\therefore y = e^{\frac{(0.2)^2}{2}} - 1 =$$

~~For, $x=0.1$, $y=1.01$~~

~~$\therefore 1.01 =$~~

$$\Rightarrow y+1 = e^{\left(\frac{x^2}{2} + c\right)}$$

$$\text{At, } x=0, y(0)=1$$

$$\Rightarrow 2 = e^c$$

$$\Rightarrow c = \ln(2)$$

$$\therefore y = e^{\frac{x^2}{2} + \ln(2)} - 1$$

$$= e^{0.713} - 1$$

$$= 1.035668$$

$$\text{Error} = |1.0402505 - 1.035668|$$

$$= 4.58 \times 10^{-3}$$

Numerical Methods:-

22/02/2023

Q. Find y at $x=0.1$ correct to three decimal places, given,
 $y' - 2y = 3e^x$, $y(0) = 0$, Using Taylor series method.

$$\hookrightarrow y_{i+1} = y_i + h y_i' + \frac{h^2}{2!} y_i''$$

$$y_1 = y_0 + h y_0' + \frac{h^2}{2!} y_0''$$

$$= 0 + h \cdot 3 + \frac{h^2}{2!} \cdot 9$$

$$= 3h + \frac{9}{2} h^2$$

$$y_2 = y_1 + h y_1' + \frac{h^2}{2!} y_1''$$

$$y' = 2y + 3e^x \quad h = x_{i+1} - x_i$$

$$y'' = 2y' + 3e^x$$

$$x_0 = 0 \quad y(0) = 0$$

x_1

$$y_0' = 2y_0 + 3e^0$$

$$= 0 + 3 = 3$$

$$y_0'' = 2y_0' + 3e^0 = 2 \cdot 3 + 3 = 9$$

$$y' = 2y + 3e^x$$

$$\therefore \frac{dy}{dx} = 2y + 3e^x$$

Modified Euler's Method and Huen's Method:-

$$\int_{x_i}^{x_{i+1}} \frac{dy}{dx} dx = \int_{x_i}^{x_{i+1}} f(x, y) dx \quad y' = f(x, y) \quad [x_i, x_{i+1}]$$

$$y(x_{i+1}) - y(x_i) = \int_{x_i}^{x_{i+1}} f(x, y) dx.$$

$$y(x_{i+1}) = y(x_i) + \int_{x_i}^{x_{i+1}} f(x, y) dx$$

Mean Value Theorem of integral calculus.

$$y(x_{i+1}) - y(x_i) = (x_{i+1} - x_i) f(x_i + \theta h, y(x_i + \theta h)) \quad 0 < \theta < 1$$

$$\Rightarrow y(x_{i+1}) = y(x_i) + h f(x_i + \theta h, y(x_i + \theta h)) \quad [\because x_{i+1} - x_i = h]$$

Case-1 $\theta = 0$, Euler's Method:-

$$y_{i+1} = y_i + h f(x_i, y_i)$$

Case-2 $\theta = 1$ Backward Euler Method:-

$$y_{i+1} = y_i + h f(x_{i+1}, y_{i+1})$$

Case-3 $\theta = \frac{1}{2}$ Mid-point method:-

$$y_{i+1} = y_i + h f\left(x_i + \frac{h}{2}, y\left(x_i + \frac{h}{2}\right)\right)$$

from Euler's method,

$$y\left(x_i + \frac{h}{2}\right) = y_i + \frac{h}{2} f(x_i, y_i)$$

$$= y_i + h f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2} f(x_i, y_i)\right) \quad [\text{Modified Euler method}]$$

Truncation Error:-

$$T.E = y(x_{i+1}) - y(x_i) - h f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2} f(x_i, y_i)\right)$$

$$= \left[y + h y' + \frac{h^2}{2!} y'' + \dots \right] - y - h \left[f + \frac{h}{2} f_x + \frac{h}{2} f_y + (\text{three terms of } h^2) + \dots \right],$$

where all terms are evaluated at x_i .

$$T.E = \left[y + h f + \frac{h^2}{2} (f_x + f_y) + \text{terms of } h^3 + \dots \right] - y - h \left[f + \frac{h}{2} f_x + \frac{h}{2} f_y + (\text{terms of } h^2) + \dots \right]$$

Truncation error is of $O(h^3) \Rightarrow$ method is second order.

Case-1 $\Rightarrow$ Approx the continuously varying slope in $[x_i, x_{i+1}]$ by mean of slopes at points x_i and x_{i+1} .

$$y_{i+1} = y_i + \frac{h}{2} [f(x_i, y_i) + f(x_{i+1}, y_{i+1})]$$

Implicit method $\Rightarrow$ Trapezium method.

$$\begin{aligned} \text{Explicit method} \Rightarrow y_{i+1} &= y_i + h f(x_i, y_i) \\ &= y_i + \frac{h}{2} \left(f(x_i, y_i) + f(x_{i+1}, y_i + h f(x_i, y_i)) \right) \end{aligned}$$

[y_{i+1} term only at one side]

Euler Cauchy or Huen's method.

Truncation error = $O(h^3) \Rightarrow$ method is second order.

Runge Kutta Methods:-

$$\int_{x_i}^{x_{i+1}} \frac{dy}{dx} dx = \int_{x_i}^{x_{i+1}} f(x, y) dx$$

$[x_i, x_{i+1}] \rightarrow$ implicit method.

In Runge Kutta methods, include slope at $x = x_i \Rightarrow \text{slope } f(x_i, y_i)$

Runge Kutta Method of 2nd order:-

Define:- Let, $k_1 = h f(x_i, y_i) = h f_i$
 $k_2 = h f(x_i + c_2 h, y_i + a_{21} k_1) = h f(x_i + c_2 h, y_i + a_{21} h f_i)$

$$y_{i+1} = y_i + w_1 k_1 + w_2 k_2$$

c_2, a_{21}, w_1, w_2 : parameters chosen such that the method is of highest possible order.

$$y(x_{i+1}) = y(x_i) + h y'(x_i) + \frac{h^2}{2!} y''(x_i) + \frac{h^3}{3!} y'''(x_i) + \dots$$

$$= y(x_i) + h f(x_i, y(x_i)) + \frac{h^2}{2!} [f_x + f_y] + \frac{h^3}{6} [f_{xx} + 2f_{xy} + f_{yy} + f_y(f_x + f_y)] x_i + \dots \quad \text{--- (1)}$$

$$\begin{aligned}\therefore k_2 &= h f(x_i + c_2 h, y_i + a_{21} h f_i) \\ &= h \left[f_i + h (c_2 f_x + a_{21} f_y) x_i + \frac{h^2}{2} (c_2^2 f_{xx} + 2c_2 a_{21} f_{xy} + a_{21}^2 f_{yy}) x_i + \dots \right]\end{aligned}$$

From ①, we have,

$$\begin{aligned}y_{i+1} &= y_i + (w_1 + w_2) h f_i + h^2 (w_2 c_2 f_x + w_2 a_{21} f_y) x_i + \frac{h^3}{2} w_2 (c_2^2 f_{xx} \\ &\quad + 2c_2 a_{21} f_{xy} + a_{21}^2 f_{yy}) x_i + \dots\end{aligned}$$

$$\therefore w_1 + w_2 = 1, \quad w_2 c_2 = \frac{1}{2}, \quad w_2 a_{21} = \frac{1}{2}$$

$$w_1 = 1 - w_2 \quad w_2 = \frac{1}{2c_2}, \quad \Rightarrow w_2 = \frac{1}{2a_{21}}$$

$$= 1 - \frac{1}{2c_2} \quad \Rightarrow a_{21} = c_2$$

$$\begin{aligned}\therefore y_{i+1} &= w_1 k_1 + w_2 k_2 \\ &= \left(1 - \frac{1}{2c_2}\right) k_1 + \frac{1}{2c_2} k_2\end{aligned}$$

$$\text{If } c_2 = 1$$

$$y_{i+1} = y_i + \frac{1}{2} k_1 + \frac{1}{2} k_2 = y_i + \frac{h}{2} f(x_i, y_i) + \frac{h}{2} f(x_i + h, y_i + h).$$

4th order - Runge kutta Method

$$y_{i+1} = y_i + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$\text{T.E} = O(h^5)$$

$$\text{where } k_1 = h f(x_i, y_i)$$

$$k_2 = h f\left(x_i + \frac{h}{2}, y_i + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_i + \frac{h}{2}, y_i + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_i + h, y_i + k_3)$$